Treatment Assignment Design Evaluation for Spatial CRTs

Simulation Study with Heterogeneous Incidence — Results Summary

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## Loading results from: sim_results_DIM_combined_20260304_195321.rds

## Loading results from: sim_results_MLE_tau_sweep_combined_20260408_191916.rds
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1 Introduction

This report summarizes a simulation study evaluating **eight treatment assignment designs** for Spatial Cluster Randomized Trials (CRTs) when outcome incidence is heterogeneous and spatially structured. The central research question is:

> *Given known historical incidence data exhibiting spatial heterogeneity, which treatment assignment strategy minimizes estimation error for the direct intervention effect $\hat{\tau}$ across a range of spatial dependence and spillover configurations?*

Application context: Sudden Unexpected Death (SUD) in North Carolina counties. The Poisson incidence mode uses an SUD base rate of 35 per 100,000 (Mirzaei et al.), consistent with county-level analyses by Gan et al. and Watson et al.

The study compares two estimation methods — Difference in Means (DIM) and Maximum Likelihood Estimation via a spatial autoregressive model (MLE). The primary MLE results use the $\tau = 1.0$ scenario (2,560 parameter scenarios); a tau sensitivity analysis across $\tau \in \{0.8, 1.0, 1.5, 2.0, 3.0\}$ (12,800 total scenarios) is presented in Section 9.

---

2 Mathematical Framework

2.1 Study Region and Spatial Structure

The simulation operates on a **regular lattice grid** $\mathcal{G} = \{s_1, \dots, s_N\}$ on $[1, G] \times [1, G]$, with $N = G^2$ clusters ($G = 10$, $N = 100$).

Two contiguity structures define neighbor relationships:

- **Rook contiguity:** $s_j \in \mathcal{N}(s_i)$ iff s_i and s_j share an edge (up to 4 neighbors)
- **Queen contiguity:** $s_j \in \mathcal{N}(s_i)$ iff s_i and s_j share an edge or corner (up to 8 neighbors)

The spatial weight matrix $\mathbf{W}$ has entries $w_{ij} = |\mathcal{N}(s_i)|^{-1}$ if $s_j \in \mathcal{N}(s_i)$ and 0 otherwise (row-standardized).

2.2 Outcome Data Generating Process

Outcomes follow a **Spatial Durbin Model (SDM)**:

$$\mathbf{Y} = (\mathbf{I} - \rho\mathbf{W})^{-1} \left[\tau\mathbf{Z} + \gamma \cdot \text{Spill}(\mathbf{Z}) + \beta\mathbf{X} + \varepsilon\right]$$

Parameter definitions:

Symbol	Role	Value / Range
τ	Direct treatment effect (estimand)	$\{0.8, 1.0, 1.5, 2.0, 3.0\}$; primary: 1.0
ρ	Outcome spatial autocorrelation	$\{0.00, 0.01, 0.20, 0.50\}$
γ	Spillover magnitude	$\{0.5, 0.6, 0.7, 0.8\}$

Symbol	Role	Value / Range
β	Incidence effect on outcome	1.0 (fixed)
σ	Residual standard deviation	1.0 (fixed)

Spillover construction:

$$\text{Spill}_i(\mathbf{Z}) = \frac{\sum_{j \in \mathcal{N}(i)} Z_j}{|\mathcal{N}(i)|}$$

Two spillover modes are evaluated: - **control_only:** γ applied only to units with $Z_i = 0$ - **both:** γ applied to all units regardless of treatment status

Stationarity requires $|\rho| < 1/\lambda_{\max}(\mathbf{W})$; all swept ρ values satisfy this condition.

2.3 Incidence Generation

Three modes generate the cluster-level covariate $X_i \in [0, 1]$, each representing a different assumption about how historical incidence arises:

Mode 1 — iid Uniform:

$$X_i \overset{\text{iid}}{\sim} \text{Uniform}(0, 1)$$

No spatial structure; serves as the baseline. One config (rho_X = N/A).

Mode 2 — Spatial (SAR filter):

$$\mathbf{X}^* = (\mathbf{I} - \rho_X \mathbf{W})^{-1} \mathbf{u}, \quad \mathbf{u} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

$$X_i = \Phi(X_i^*) \in (0, 1)$$

Spatially correlated via SAR filter; probability integral transform (Φ) ensures X_i is marginally Uniform$(0, 1)$ but spatially dependent. Two configs: $\rho_X \in \{0.20, 0.50\}$.

Mode 3 — Poisson (count rates):

1. Spatially correlated log-rates: $\log \lambda_i^* = \mu_0 + [(\mathbf{I} - \rho_X \mathbf{W})^{-1} \mathbf{u}]_i$
2. Poisson counts: $C_i \sim \text{Poisson}(\lambda_i^* \cdot P_i)$ where $P_i = 1{,}000$ (equal population per cluster)
3. Observed rates: $R_i = C_i / P_i$
4. Rank-normalized: $X_i = (\text{rank}(R_i) - 0.5)/N$

Base rate $\mu_0 = \log(35/100{,}000)$ mimics SUD incidence. Two configs: $\rho_X \in \{0.20, 0.50\}$.

Critical design invariant: Incidence is generated **once per (mode, ρ_X) configuration** and held fixed across all (ρ, γ, nb_type) scenarios. This reflects the realistic constraint that a researcher observes historical incidence once before designing the trial; spatial model parameters affect the outcome DGP, not the incidence itself.

2.4 Treatment Assignment Designs

Six assignment strategies are evaluated, representing qualitatively distinct approaches to allocating approximately 50% of clusters to treatment:

ID	Name	Deterministic?	Key Feature
1	Checkerboard	Yes	Alternating grid; maximizes spatial separation
2	High Incidence Focus	Yes	Treats top-50% incidence clusters
3	Saturation Quadrants	No	Random saturation level per quadrant (0.2–0.8)
4	Isolation Buffer	No	Greedy assignment; no adjacent treated clusters
5	$2\$\times 2 Blocking \| No \| 1 : 1 randomization within 2\times\$2$ spatial blocks		
7	Balanced Quartiles	No	Stratified by incidence quartile, balanced within strata

Designs 1 and 2 are **deterministic** given the incidence vector: they produce the same assignment on every resample. The simulation detects this and runs each once, replicating the result across all design resamples (saving $ $25x compute).

2.5 Estimation Methods

Difference in Means (DIM):

$$\hat{\tau}_{\text{DIM}} = \bar{Y}_1 - \bar{Y}_0 = \frac{1}{n_1}\sum_{Z_i=1} Y_i - \frac{1}{n_0}\sum_{Z_i=0} Y_i$$

95% CI via Neyman variance estimator:

$$\widehat{\text{Var}}(\hat{\tau}) = \frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}$$

Maximum Likelihood (MLE via SAR):

Fits `lagsarlm(Y ~ Z + Spill + X, listw = W)` from the `spatialreg` package, simultaneously estimating ρ, τ, and regression coefficients. The true spillover covariate Spill($\mathbf{Z}$) is included as an **oracle regressor** (i.e., the researcher is assumed to know the true spillover structure). CIs are derived from the asymptotic covariance matrix.

---

3 Simulation Algorithm

3.1 Parameter Grid

Table 1: Simulation parameter grid (1,920 total scenarios per estimator)

Parameter	Values	Levels

Incidence mode	iid, spatial, poisson	5 configs (1 iid + 2 spatial + 2 poisson)
rho_X (incidence spatial autocorr.)	N/A (iid); 0.20, 0.50 (spatial, poisson)	—
nb_type (contiguity)	rook, queen	2
rho (outcome spatial autocorr.)	0.00, 0.01, 0.20, 0.50	4
gamma (spillover magnitude)	0.5, 0.6, 0.7, 0.8	4
spill_type	control_only, both	2
design_id	1, 2, 3, 4, 5, 7	6
TOTAL		**1,920 scenarios**

3.2 Resampling Structure

For each scenario, the simulation draws:

Mode	Design resamples (R_d)	Outcome resamples (R_o)	Iterations / scenario
DIM	25	100	2,500
MLE	25	10	250

3.3 Evaluation Metrics

For each scenario, across K total iterations (design $\times$ outcome resamples):

$$\text{Bias} = \bar{\hat{\tau}} - \tau, \quad \text{SD} = \sqrt{\frac{1}{K-1}\sum_{k=1}^{K}(\hat{\tau}_k - \bar{\hat{\tau}})^2}$$

$$\text{MSE} = \text{Bias}^2 + \text{SD}^2$$

$$\text{Coverage} = \frac{1}{K}\sum_{k=1}^{K}\mathbf{1}\left[\tau \in (\hat{\tau}_{k,L}, \hat{\tau}_{k,U})\right]$$

3.4 Per-Scenario Reproducibility

Each scenario uses a deterministic seed:

$$\text{seed}_s = \texttt{digest::digest2int}(\texttt{paste}(\text{inc_mode}, \rho_X, \text{nb_type}, \rho, \gamma, \text{spill_type}, d_{\text{id}}, \text{sep} = \text{"|"}))$$

This ensures results are reproducible and order-independent: adding or removing scenarios from the parameter sweep does not affect other scenarios' seeds.

---

4 Results: Difference in Means (DIM)

4.1 Overall Summary (All DIM Scenarios)

Table 2: DIM: Summary statistics across all 1920 scenarios

Metric	Min	Mean	Max
Bias	-0.883	-0.204	0.558
MSE	0.032	0.152	0.799
Coverage	0.000	0.720	0.990

4.2 DIM Design Comparison by MSE

Table 3: DIM: Design performance averaged across all scenarios and incidence modes

Design	Mean Bias	Median MSE	Mean MSE	Mean Coverage
Design 7	**-0.128**	**0.091**	**0.095**	**0.821**
Design 2	-0.076	0.081	0.101	0.822
Design 3	-0.011	0.090	0.111	0.802
Design 5	-0.253	0.101	0.128	0.736
Design 4	-0.310	0.133	0.163	0.702
Design 1	-0.448	0.253	0.314	0.446

4.3 DIM: MSE by Design and Neighbor Type

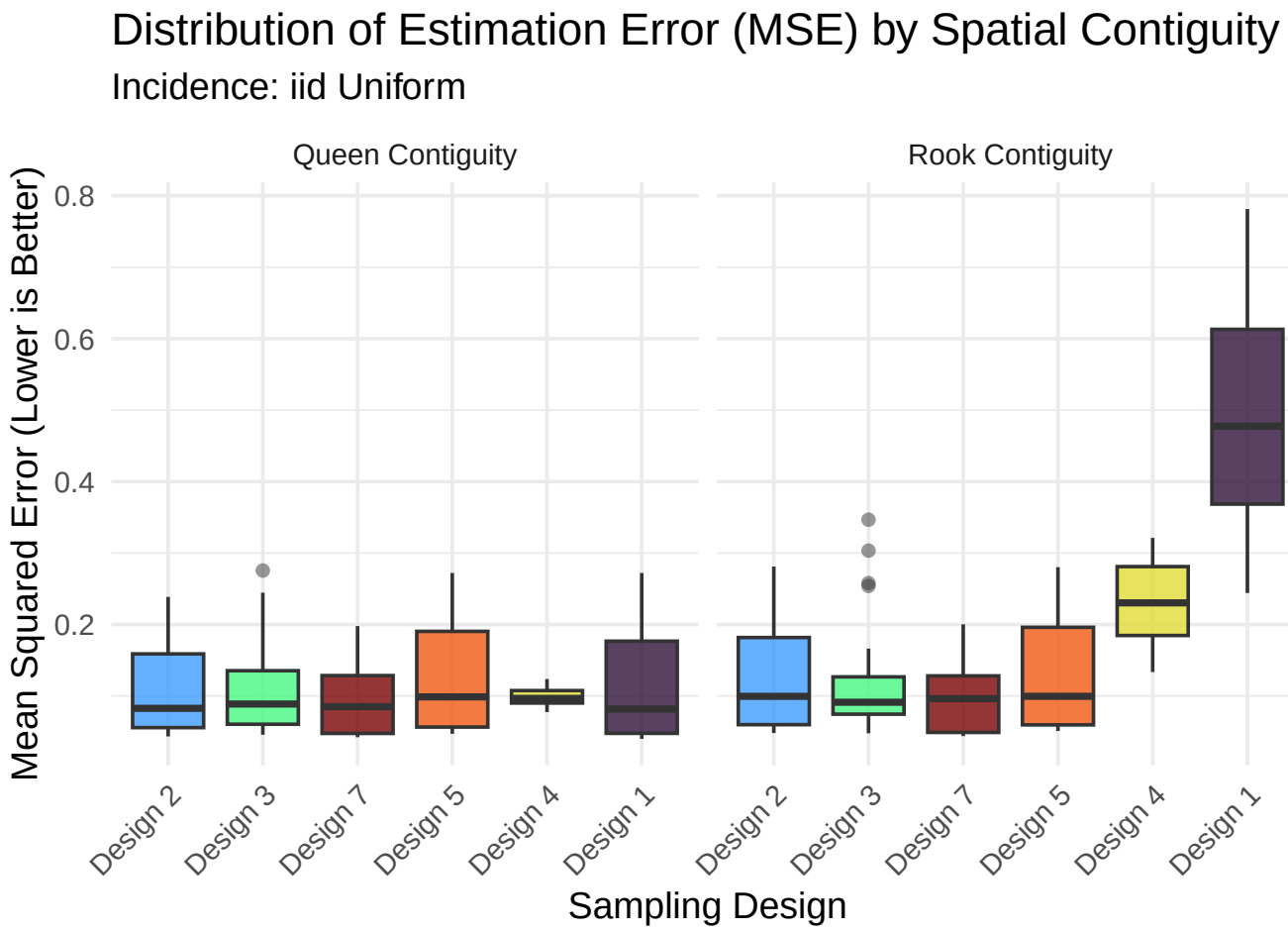

Figure 1: DIM: Distribution of MSE by design, faceted by spatial contiguity structure. Boxes sorted by median MSE within each panel.

4.4 DIM: MSE Heatmap (Rho x Design)

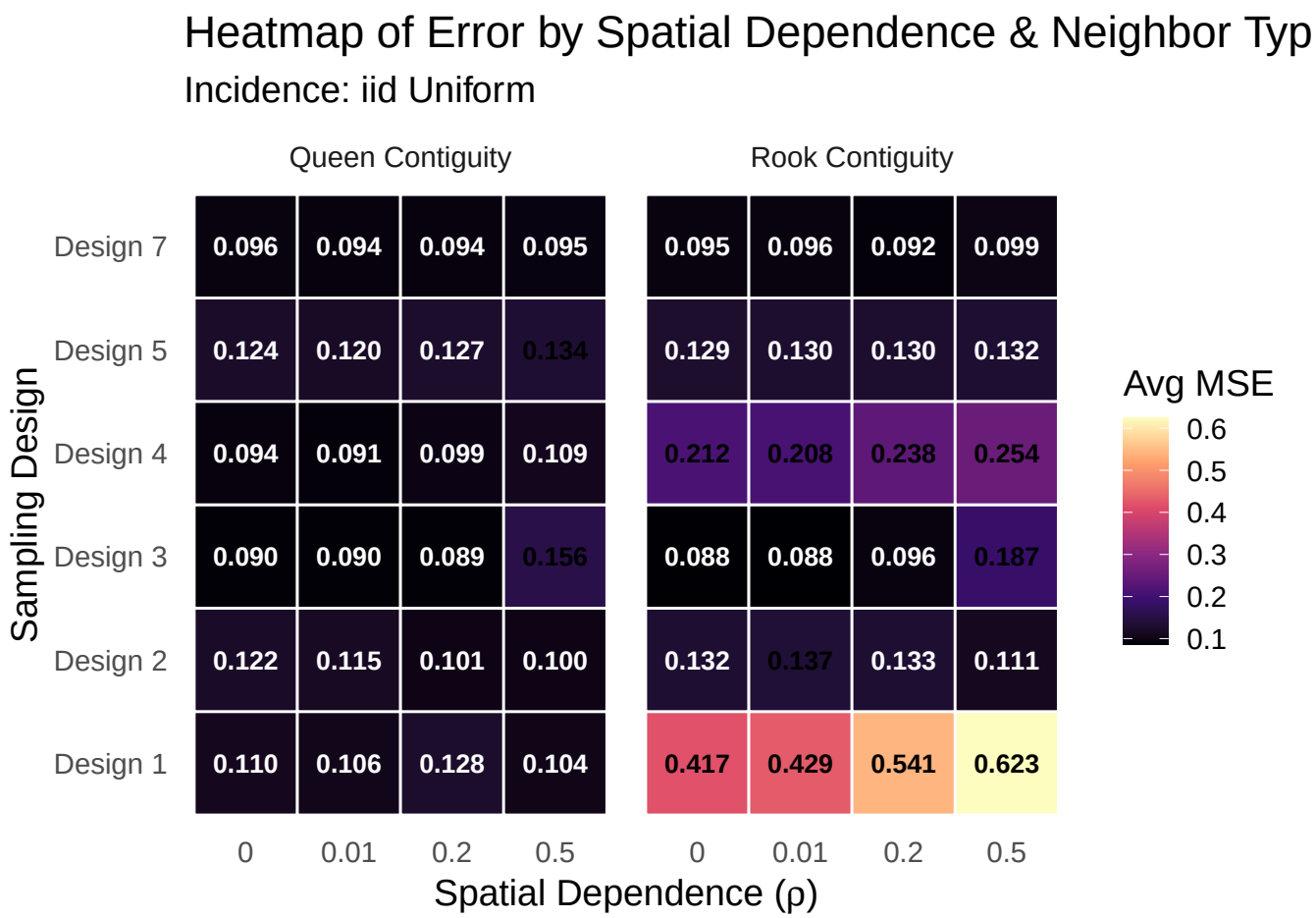

Figure 2: DIM MSE heatmap for iid Uniform incidence. Cells show average MSE across spillover type and gamma, faceted by neighbor type.

4.5 DIM: Design Rank Frequency

Table 4: DIM: Design rank frequency (%) for iid Uniform incidence. Each cell = % of scenarios where design achieved that rank (Rank 1 = lowest MSE).

Design	Rank_1	Rank_2	Rank_3	Rank_4	Rank_5	Rank_6
Design 3	32.8	29.7	7.8	9.4	14.1	6.2
Design 7	28.1	34.4	31.2	6.2	0.0	0.0
Design 4	15.6	7.8	4.7	4.7	48.4	18.8
Design 2	12.5	4.7	21.9	39.1	20.3	1.6
Design 1	10.9	7.8	4.7	14.1	6.2	56.2
Design 5	0.0	15.6	29.7	26.6	10.9	17.2

4.6 DIM: Robustness Profile

Table 5: DIM: Robustness profile for iid Uniform incidence. Sorted by Median MSE. IQR = interquartile range across scenarios.

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 2	0.0434	0.0574	0.0897	0.1754	0.2812	0.1180
Design 3	0.0458	0.0605	0.0904	0.1306	0.3467	0.0701
Design 7	0.0423	0.0478	0.0945	0.1284	0.2003	0.0806
Design 5	0.0472	0.0575	0.0996	0.1933	0.2803	0.1358
Design 4	0.0775	0.0969	0.1287	0.2295	0.3213	0.1326
Design 1	0.0401	0.0962	0.2441	0.4764	0.7815	0.3801

4.7 DIM: Coverage by Design

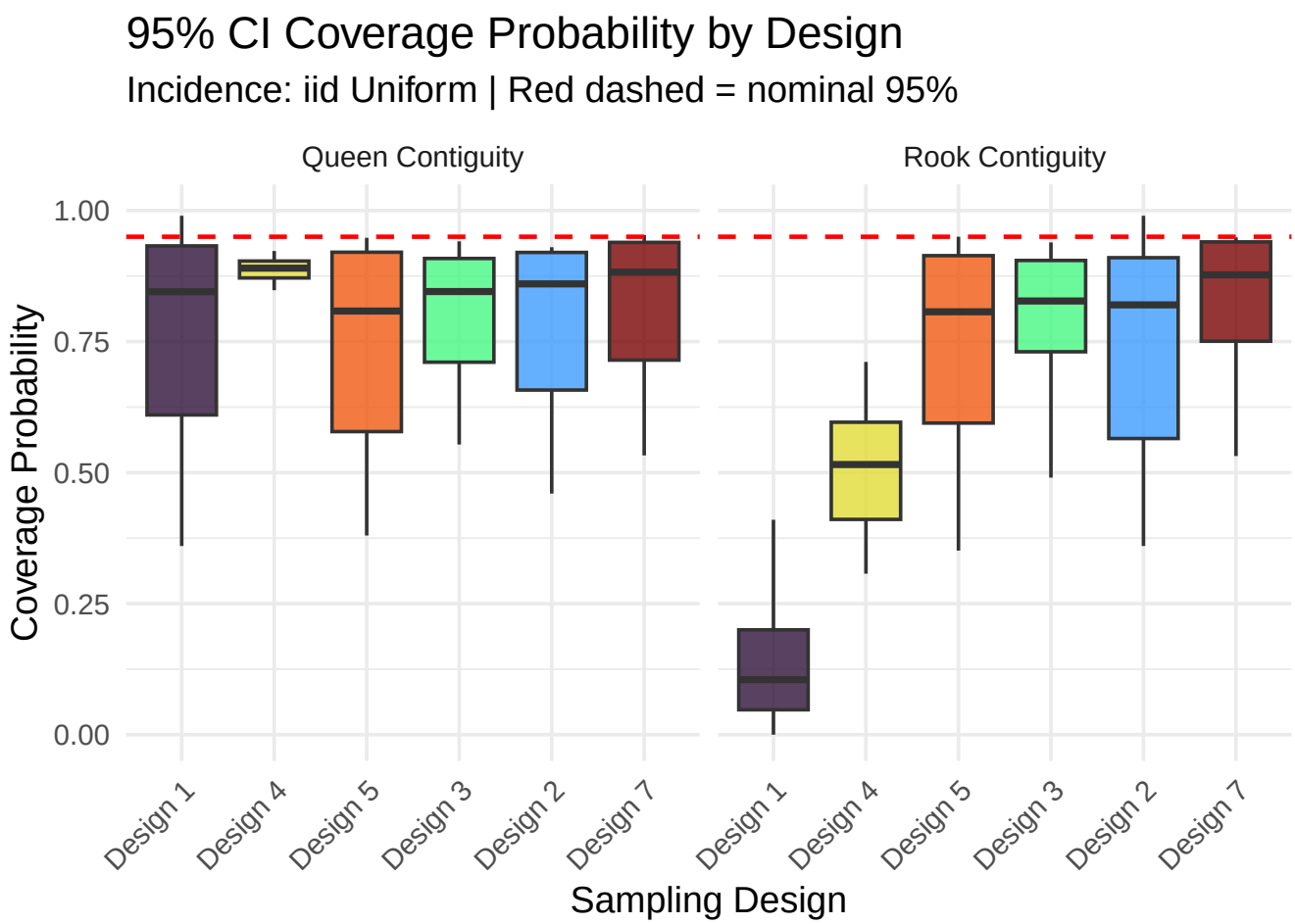

Figure 3: DIM: Coverage probability distribution by design across all scenarios. Red dashed line = nominal 95% target.

4.8 DIM: Bias-Variance Decomposition

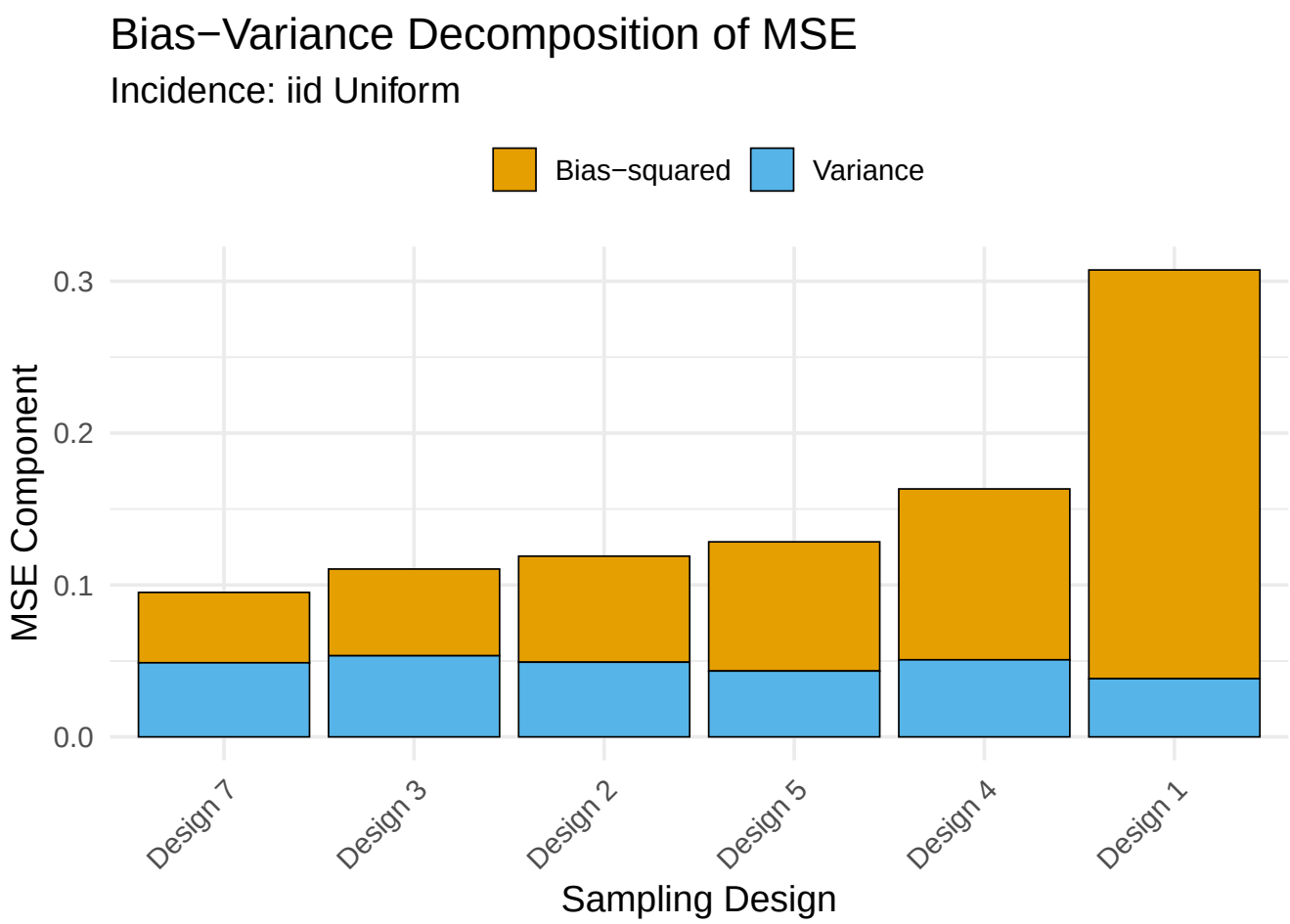

Figure 4: DIM: Bias-squared (dark) and variance (light) components of MSE per design for iid Uniform incidence. Total bar height = MSE.

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5 Results: Maximum Likelihood (MLE)

5.1 Overall Summary (All MLE Scenarios)

Table 6: MLE: Summary statistics across all 2560 scenarios

Metric	Min	Mean	Max
Bias	-0.965	-0.026	0.811
MSE	0.012	0.210	4.812
Coverage	0.000	0.890	1.000
Fail Rate	NA	0.000	NA

5.2 MLE Design Comparison by MSE

Table 7: MLE: Design performance averaged across all scenarios and incidence modes

Design	Mean Bias	Median MSE	Mean MSE	Mean Coverage	Fail Rate
Design 8	**0.025**	**0.059**	**0.079**	**0.941**	**0**
Design 3	0.016	0.065	0.080	0.939	0
Design 6	0.010	0.076	0.102	0.938	0
Design 7	0.012	0.079	0.110	0.937	0
Design 2	0.013	0.086	0.133	0.966	0
Design 4	0.012	0.146	0.148	0.939	0
Design 5	0.007	0.120	0.228	0.937	0
Design 1	-0.305	0.456	0.802	0.527	0

5.3 MLE: MSE by Design and Neighbor Type

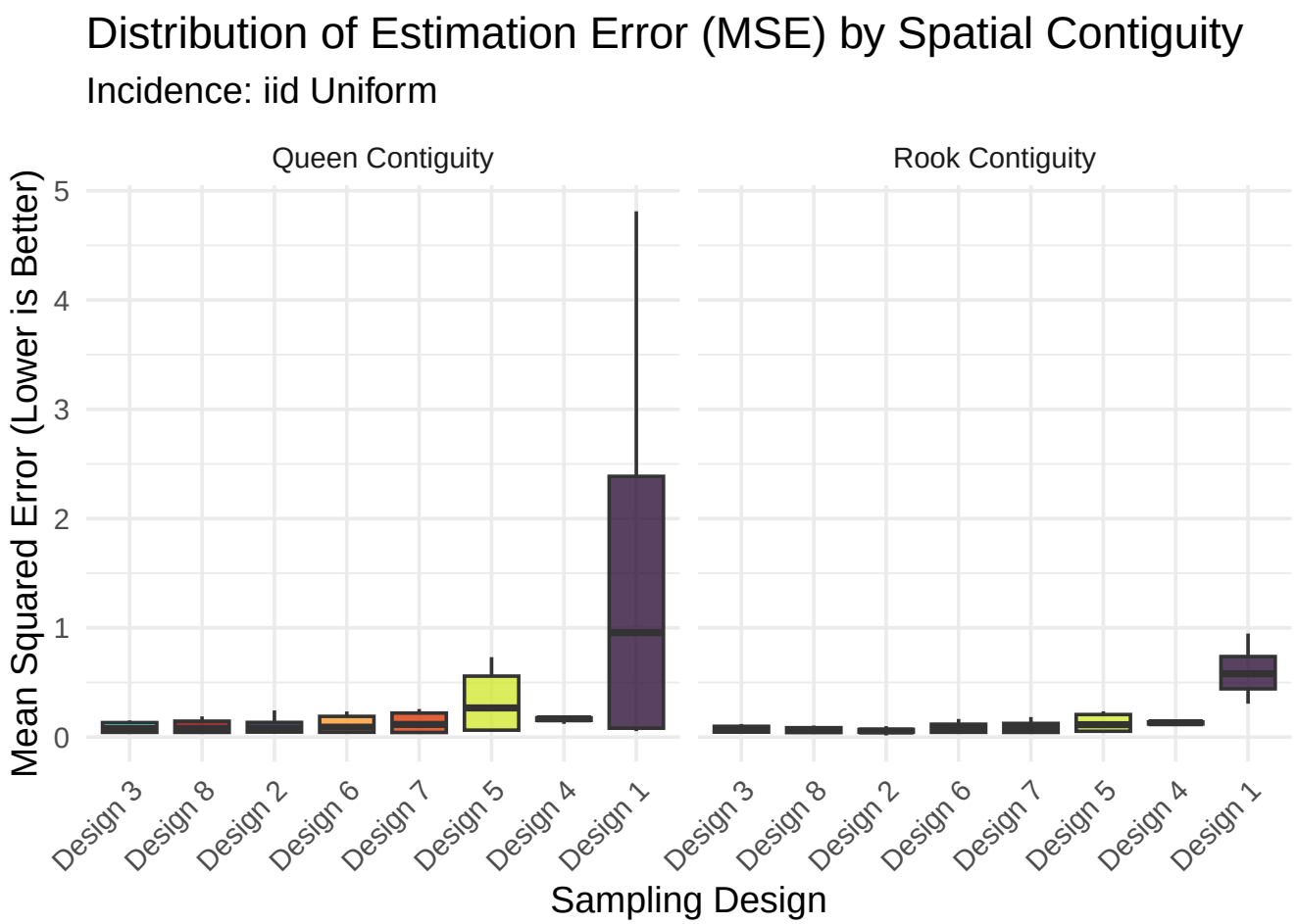

Figure 5: MLE: Distribution of MSE by design, faceted by spatial contiguity. Note: high-rho scenarios produce inflated MSE for some designs due to reduced iteration count (250 vs 2,500 for DIM).

5.4 MLE: MSE Heatmap

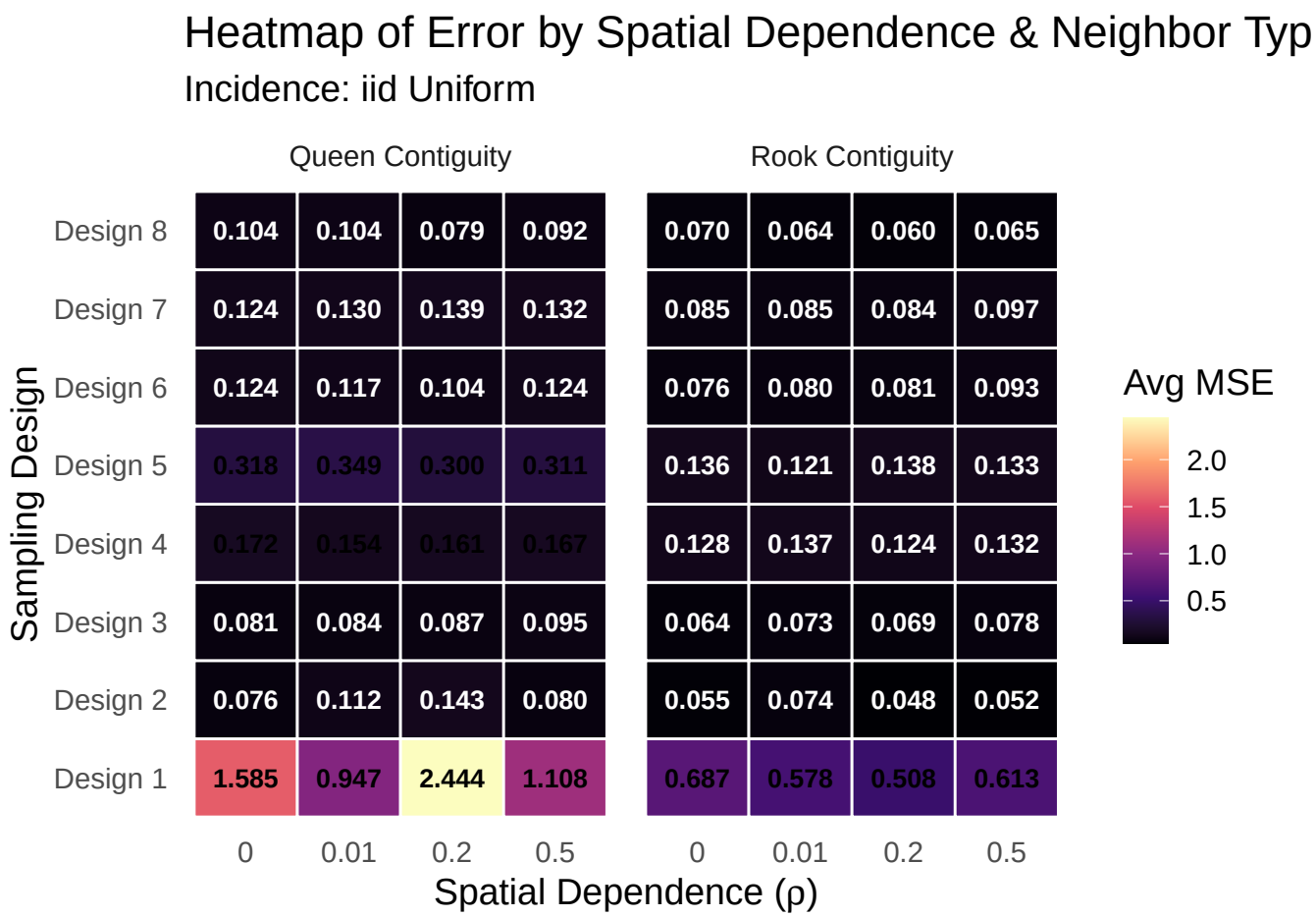

Figure 6: MLE MSE heatmap for iid Uniform incidence. Elevated cells at rho = 0.50 reflect variance inflation with fewer iterations.

5.5 MLE: Design Rank Frequency

Table 8: MLE: Design rank frequency (%) for iid Uniform incidence.

Design	Rank_6	Rank_7	Rank_8	Rank_1	Rank_2	Rank_3	Rank_4	Rank_5
Design 2	6.2	0.0	0	45.3	9.4	12.5	10.9	15.6
Design 8	0.0	0.0	0	21.9	32.8	26.6	12.5	6.2
Design 3	0.0	0.0	0	17.2	26.6	31.2	12.5	12.5
Design 6	7.8	0.0	0	9.4	10.9	10.9	29.7	31.2
Design 7	28.1	0.0	0	6.2	15.6	12.5	10.9	26.6
Design 1	4.7	20.3	75	0.0	0.0	0.0	0.0	0.0
Design 4	12.5	25.0	25	0.0	4.7	4.7	21.9	6.2
Design 5	40.6	54.7	0	0.0	0.0	1.6	1.6	1.6

5.6 MLE: Robustness Profile

Table 9: MLE: Robustness profile for iid Uniform incidence.

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 3	0.0333	0.0439	0.0631	0.1077	0.1527	0.0637
Design 8	0.0352	0.0431	0.0638	0.1024	0.1894	0.0593
Design 2	0.0151	0.0453	0.0638	0.1011	0.2443	0.0558
Design 6	0.0378	0.0440	0.0783	0.1434	0.2348	0.0995
Design 7	0.0365	0.0431	0.0841	0.1841	0.2572	0.1410
Design 5	0.0464	0.0573	0.1180	0.2934	0.7316	0.2360
Design 4	0.1031	0.1309	0.1427	0.1675	0.1864	0.0366
Design 1	0.0552	0.2573	0.5803	1.1608	4.8117	0.9034

5.7 MLE: Coverage by Design

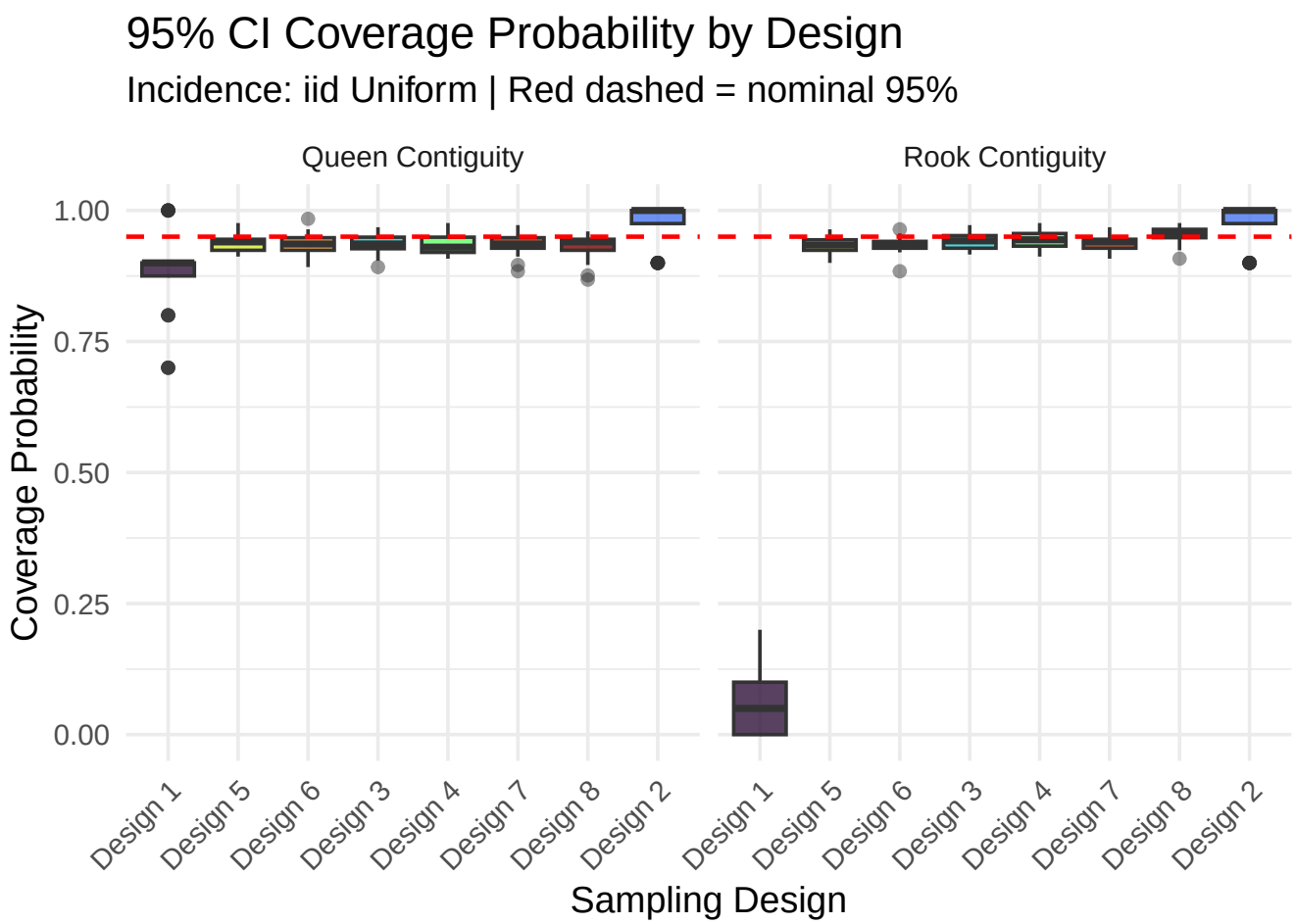

Figure 7: MLE: Coverage probability distribution by design. MLE coverage is substantially higher than DIM due to proper spatial autocorrelation modeling. Red dashed line = 95%.

6 DIM vs. MLE Comparison

6.1 Side-by-Side Metrics

Table 10: DIM vs. MLE: Average MSE and coverage by design (averaged across all scenarios and incidence modes). Results are always reported separately by incidence mode in subgroup analyses.

	DIM		MLE	
Design	DIM MSE	DIM Coverage	MLE MSE	MLE Coverage
Design 7	**0.095**	**0.821**	**0.110**	**0.937**
Design 2	0.101	0.822	0.133	0.966
Design 3	0.111	0.802	0.080	0.939
Design 5	0.128	0.736	0.228	0.937
Design 4	0.163	0.702	0.148	0.939
Design 1	0.314	0.446	0.802	0.527

6.2 Coverage Comparison by Estimator

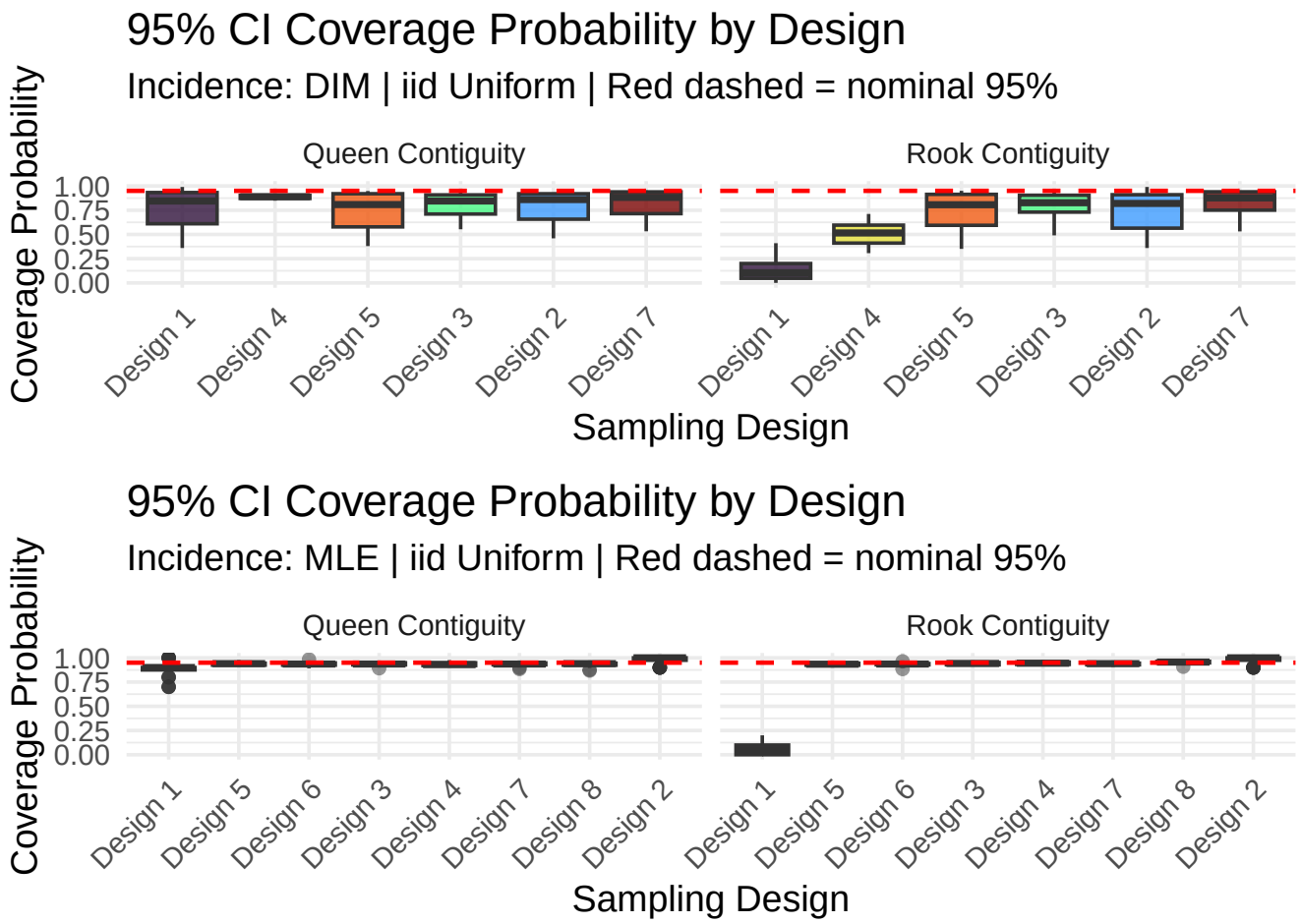

Figure 8: Coverage comparison: DIM (left) vs MLE (right) for the iid Uniform incidence configuration. MLE achieves near-nominal coverage for all designs except Checkerboard; DIM coverage is consistently below 0.95.

6.3 Key Observations

- **Coverage:** MLE achieves near-nominal 95% coverage for Designs 2–7, dramatically outperforming DIM (mean 0.72 across all scenarios). This reflects MLE's explicit modeling of spatial autocorrelation ρ, which inflates DIM standard errors.
- **MSE:** DIM median MSE is lower than MLE for most designs because DIM uses 2,500 iterations vs. 250 for MLE. MLE's maximum MSE (3.910) exceeds DIM's (0.799), occurring in high-ρ scenarios where spatial lag model variance inflates with fewer iterations.
- **Design 1 (Checkerboard):** Consistently the worst design under both estimators. Despite maximizing spatial separation, alternating treatment assignment creates a strong correlation between treatment and spatially-structured incidence, inducing systematic confounding.
- **Design 7 (Balanced Quartiles):** Top DIM performer. Stratifying by incidence quartile and randomizing within strata controls confounding from X directly, yielding low MSE and the best DIM coverage.

- **Design 3 (Saturation Quadrants):** Top MLE performer. Random saturation per quadrant introduces sufficient variation in spatial treatment patterns, allowing the spatial autoregressive model to cleanly separate τ from ρ.

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7 Sensitivity Analysis

7.1 Parameter Sensitivity (DIM)

Table 11: DIM: Eta-squared (proportion of MSE variance explained) by parameter, for iid Uniform incidence. Higher values indicate parameters where design choice matters more.

Design	Rho	Gamma	Spillover_Type	Neighbor_Type
Design 1	0.0350	0.1088	0.0202	0.7220
Design 2	0.0147	0.1018	0.7598	0.0172
Design 3	0.2809	0.0582	0.0005	0.0042
Design 4	0.0274	0.1661	0.0001	0.7196
Design 5	0.0015	0.1172	0.7968	0.0006
Design 7	0.0011	0.1200	0.6661	0.0001

7.2 Effect of Rho on Design Rankings (DIM)

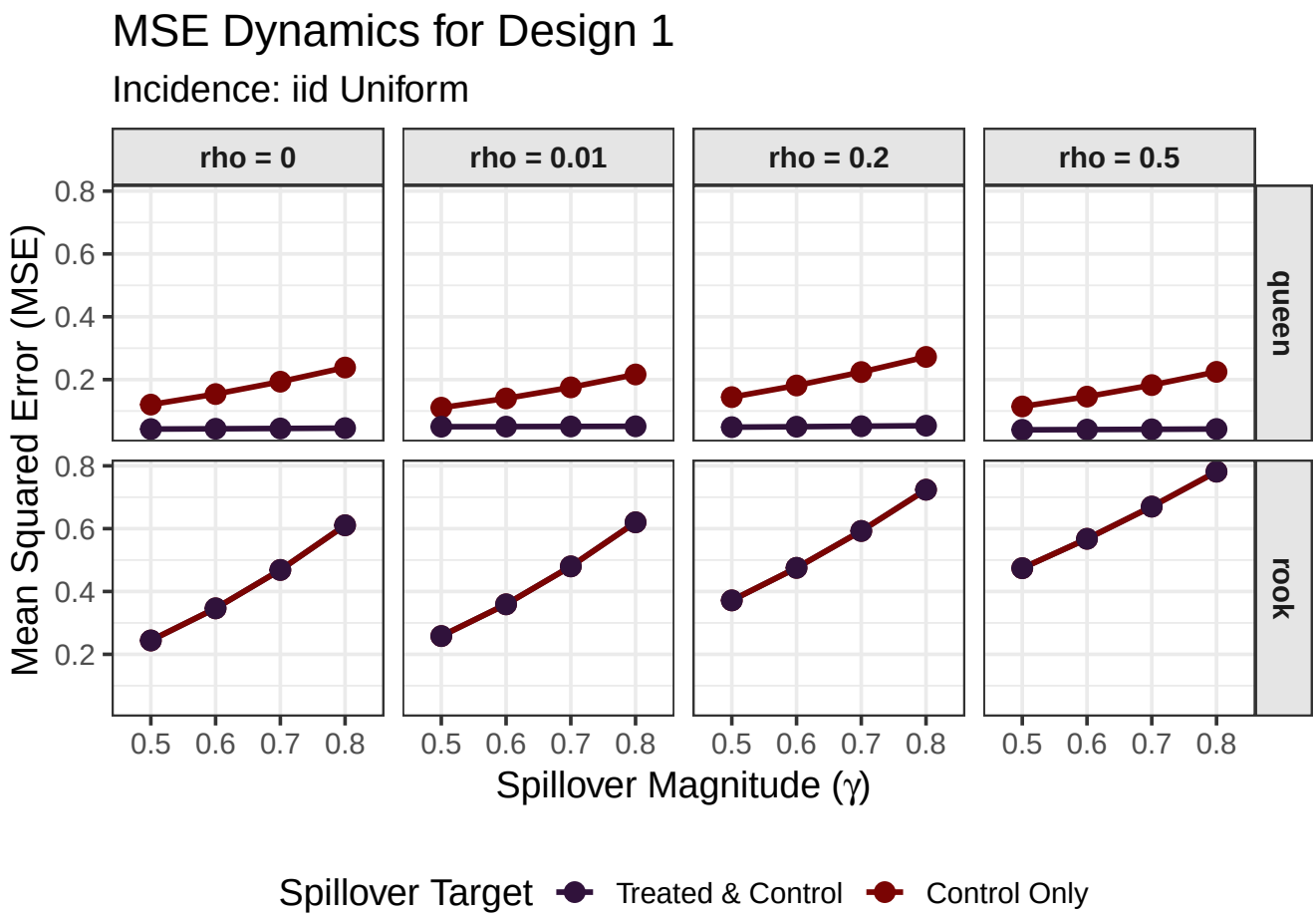
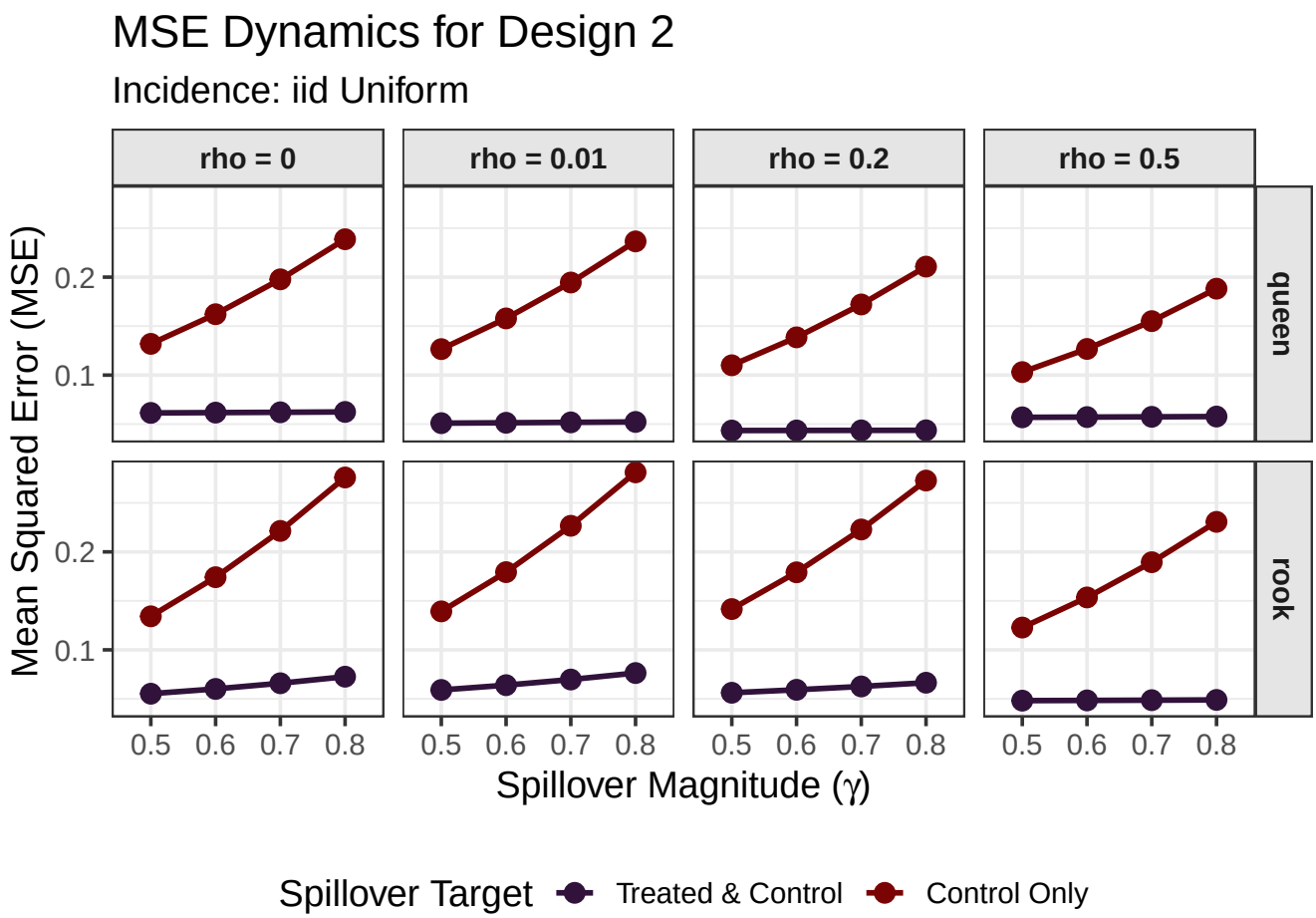


Figure 9: DIM: Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.



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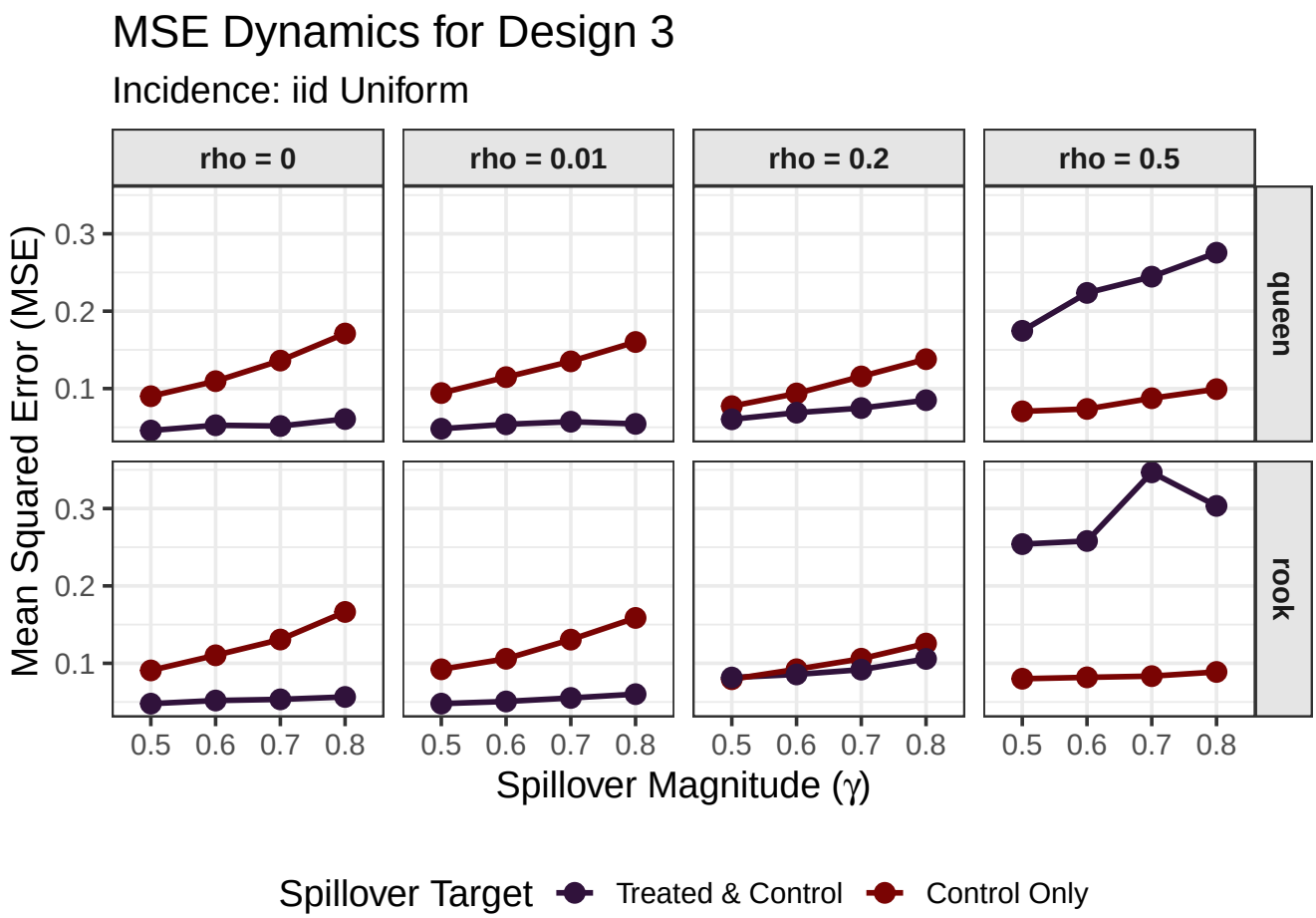

Figure 11: DIM: Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

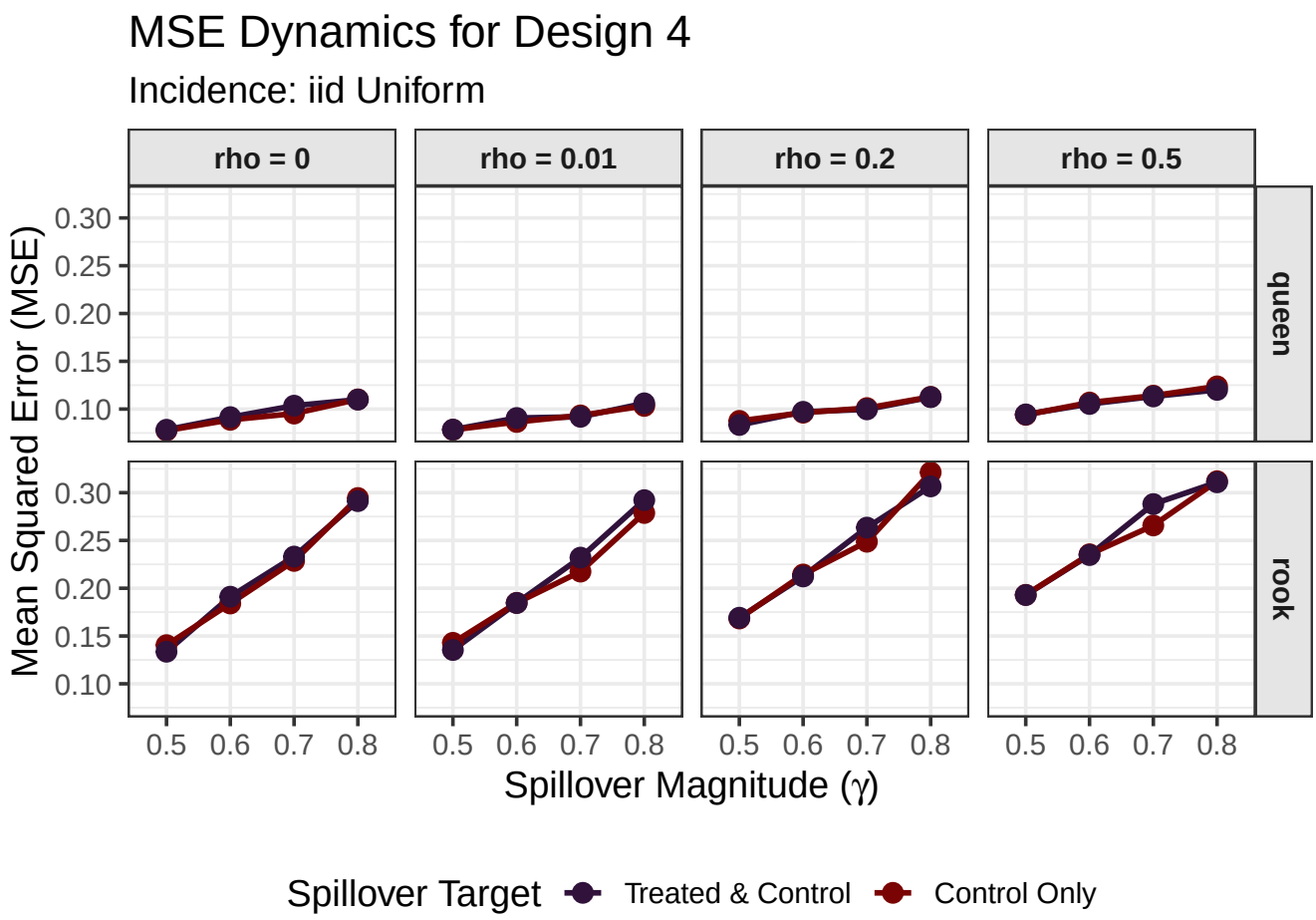

Figure 12: DIM: Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

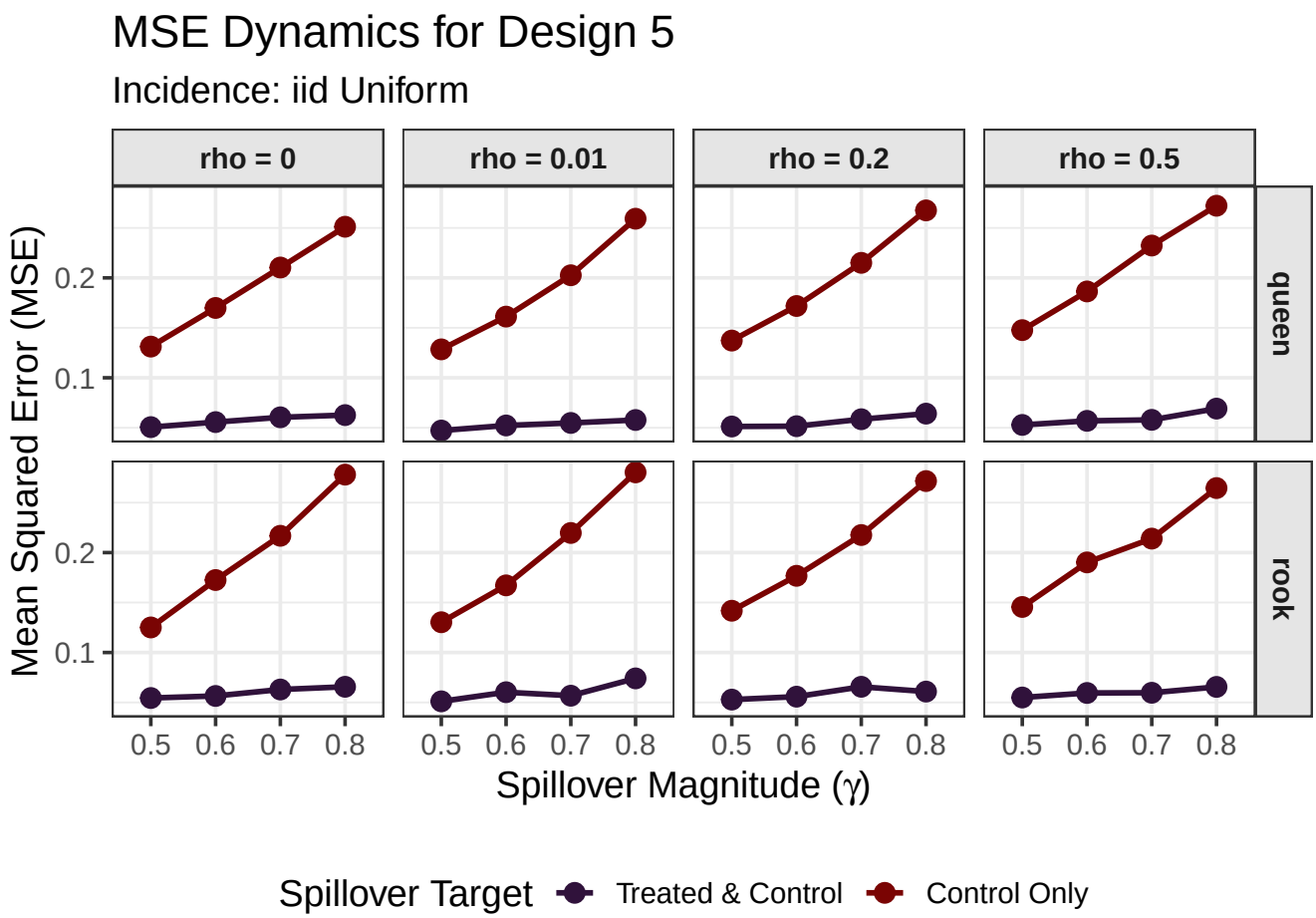

Figure 13: DIM: Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

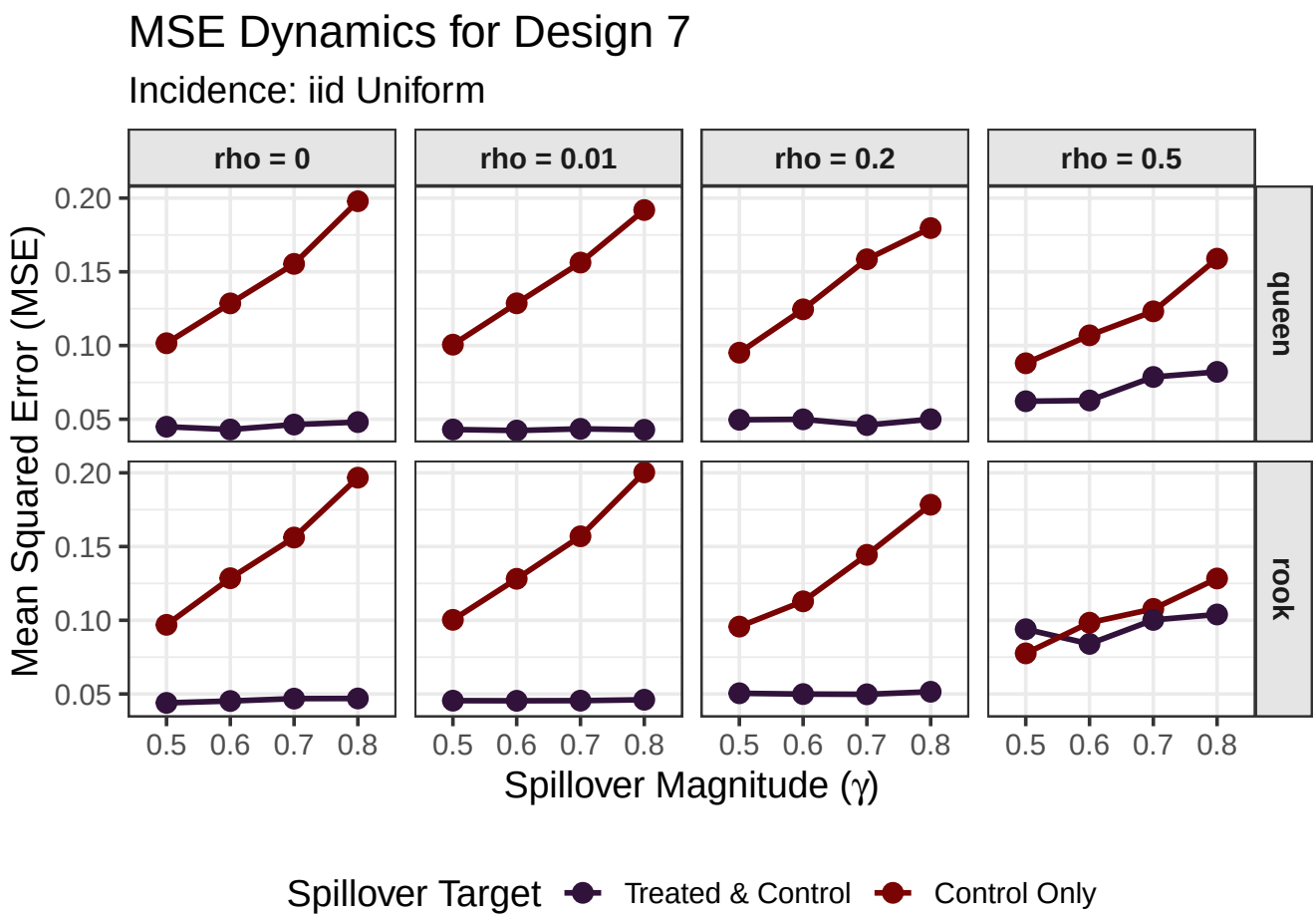

Figure 14: DIM: Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

```
## $`Design 1`
```

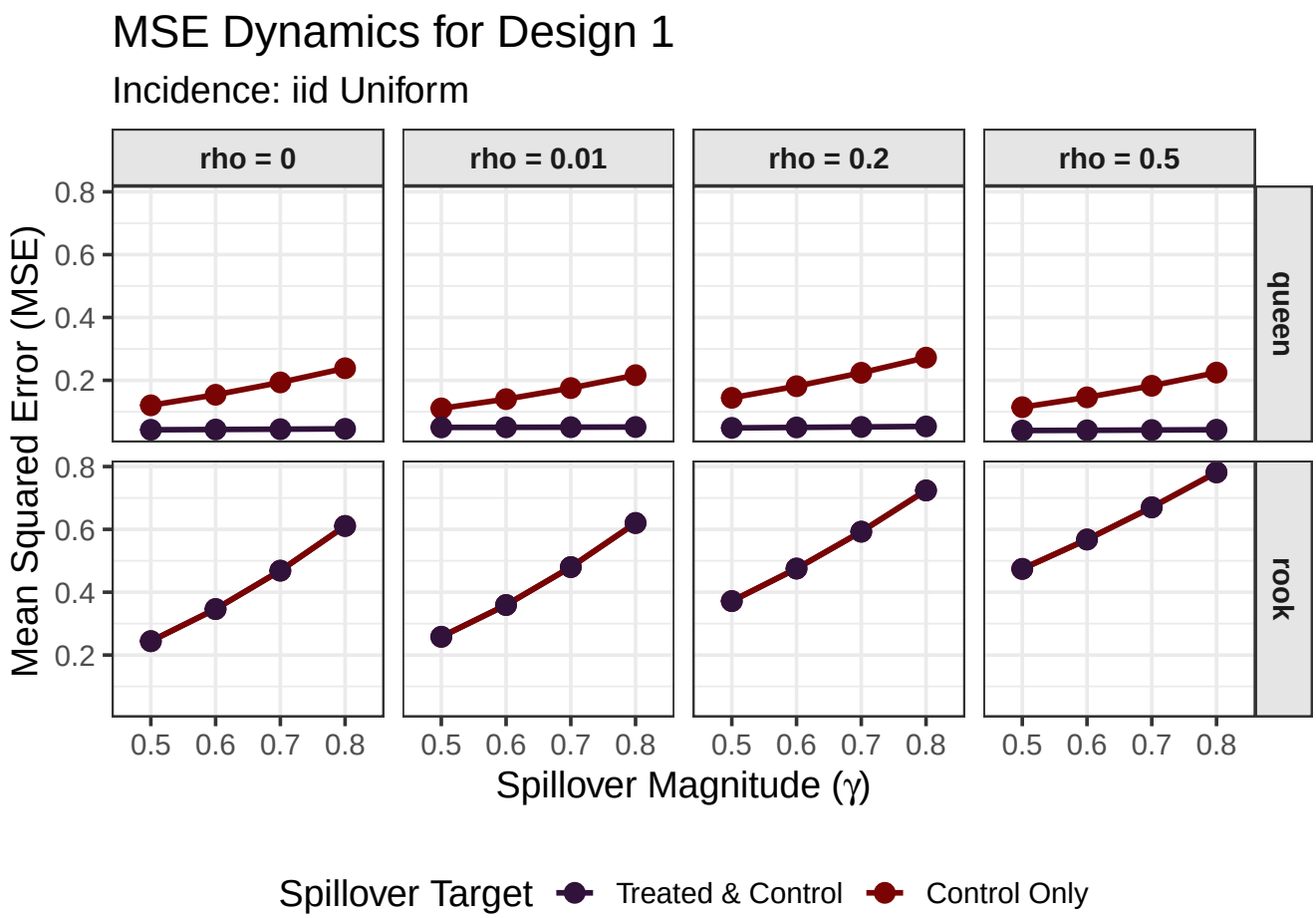

Figure 15: DIM: Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

```
##
## $`Design 2`
```

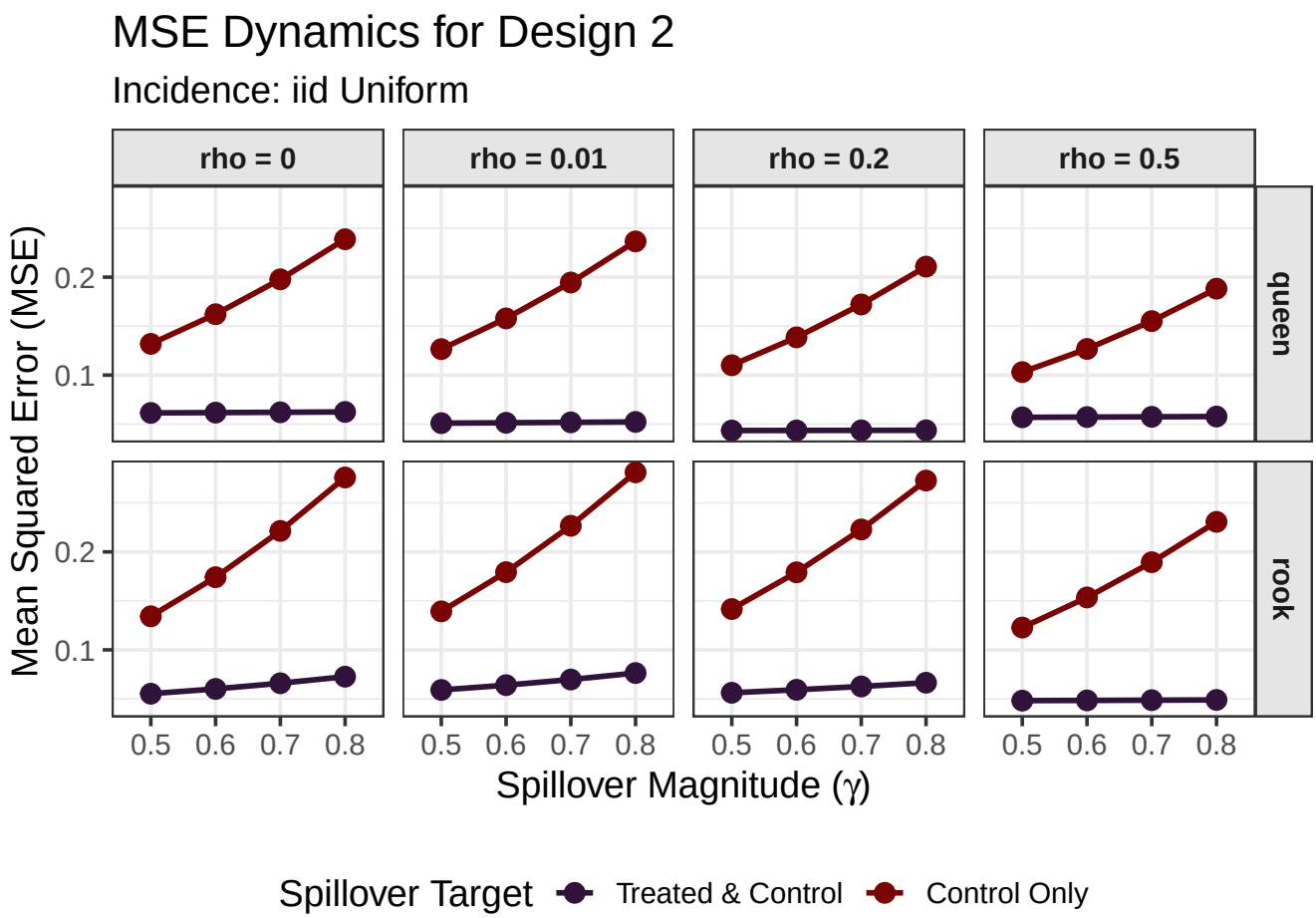

Figure 16: DIM: Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

```
##
## $`Design 3`
```

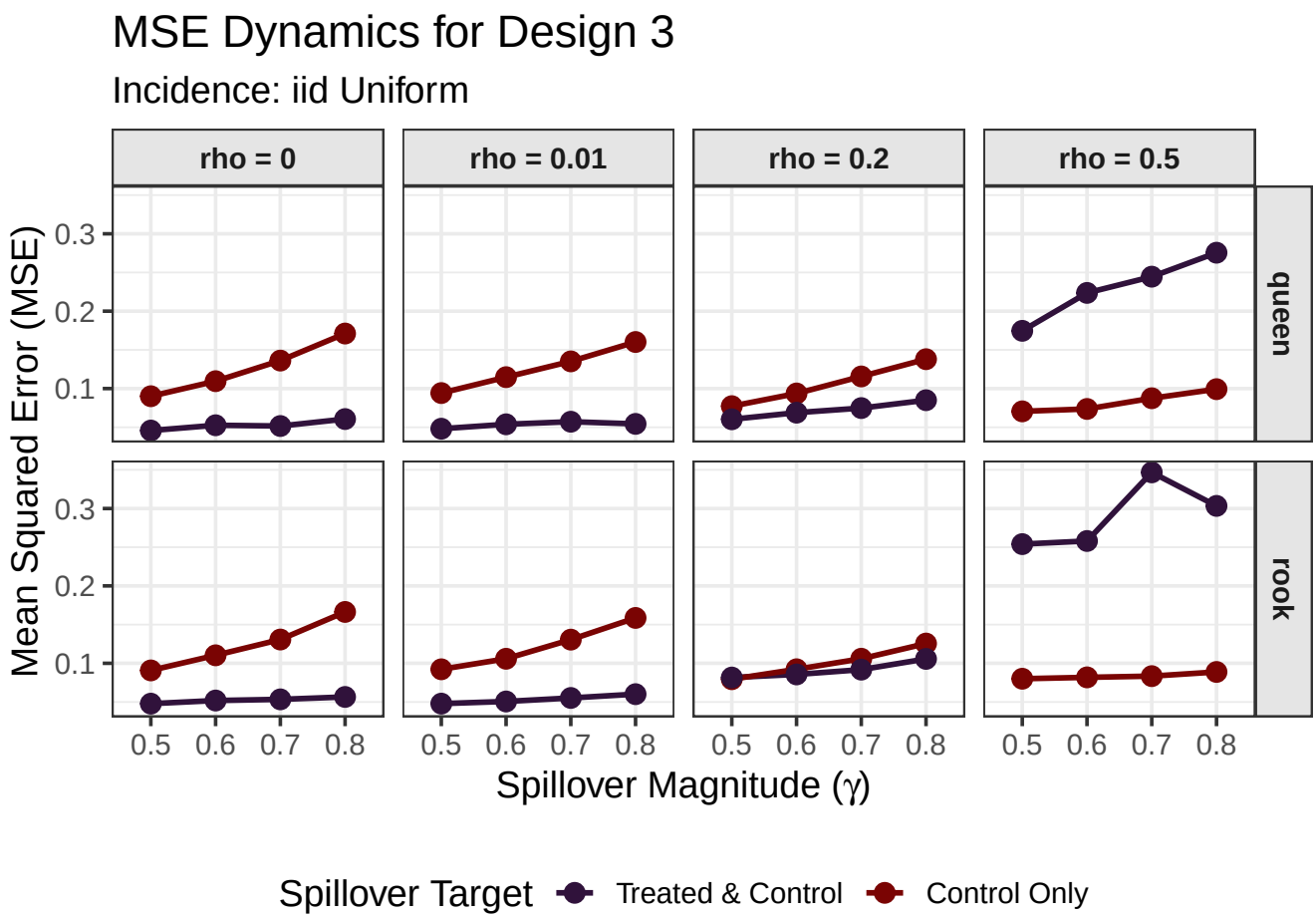

Figure 17: DIM: Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

```
##
## $`Design 4`
```

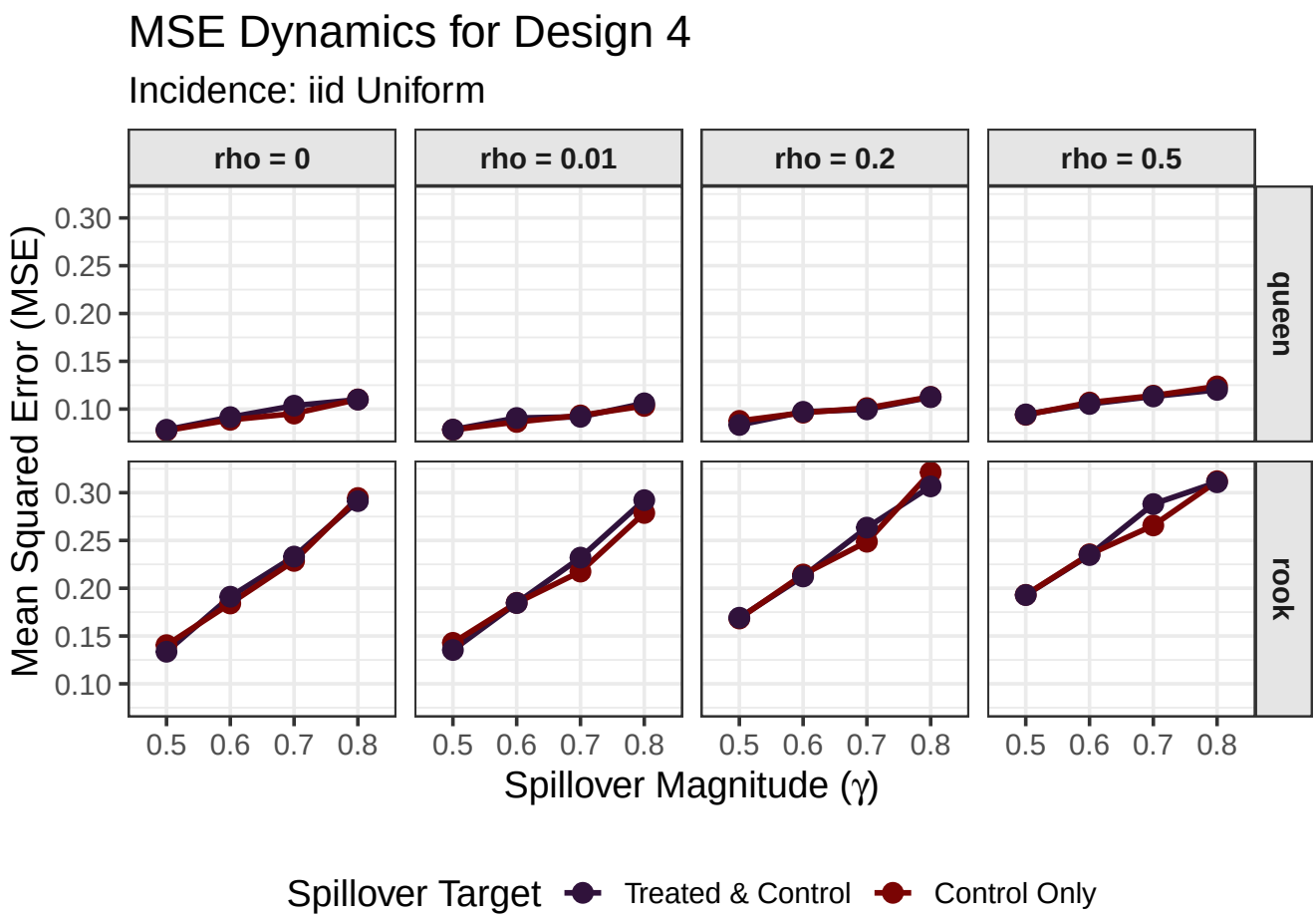

Figure 18: DIM: Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

```
##
## $`Design 5`
```

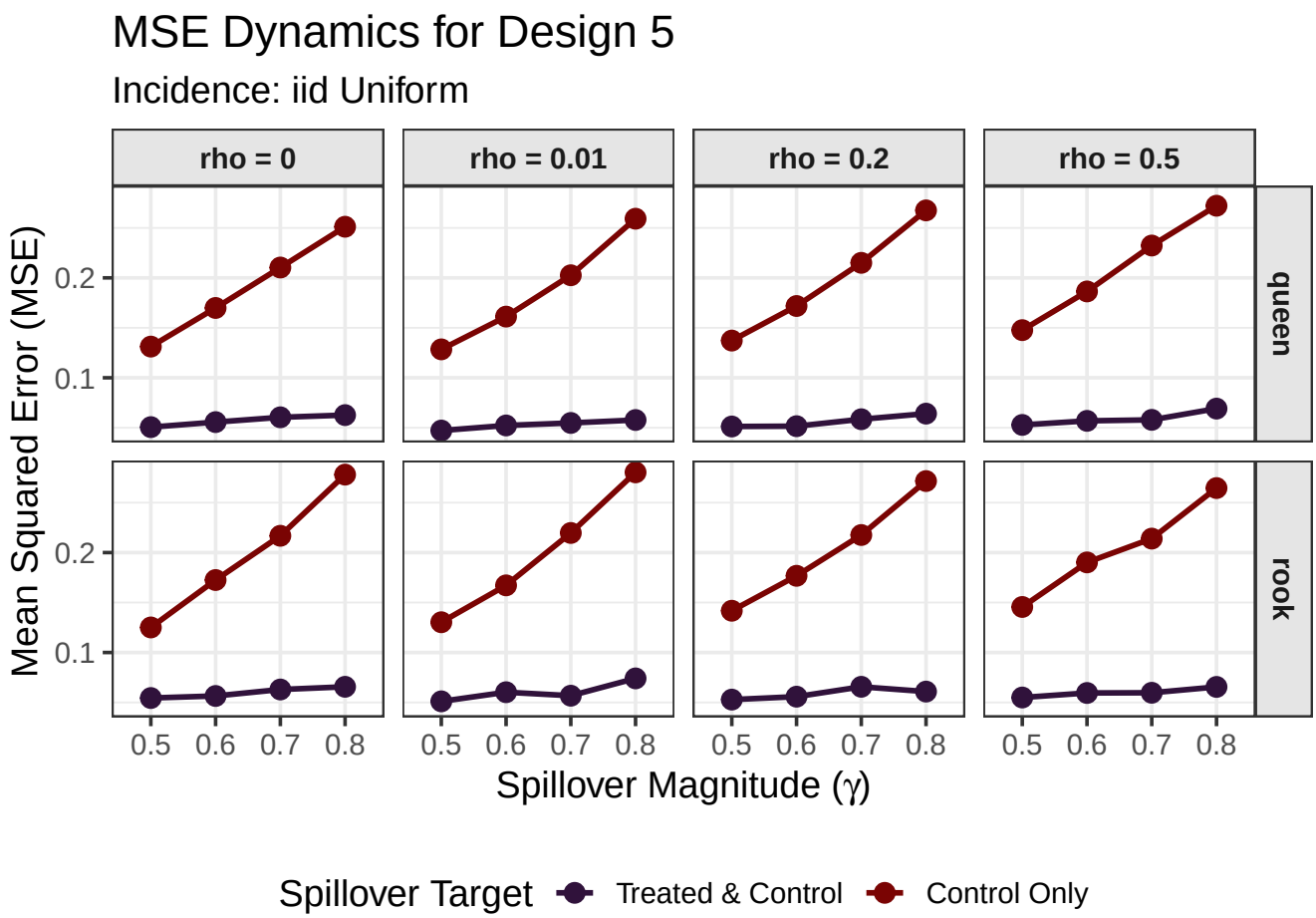

Figure 19: DIM: Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

```
##
## $`Design 7`
```

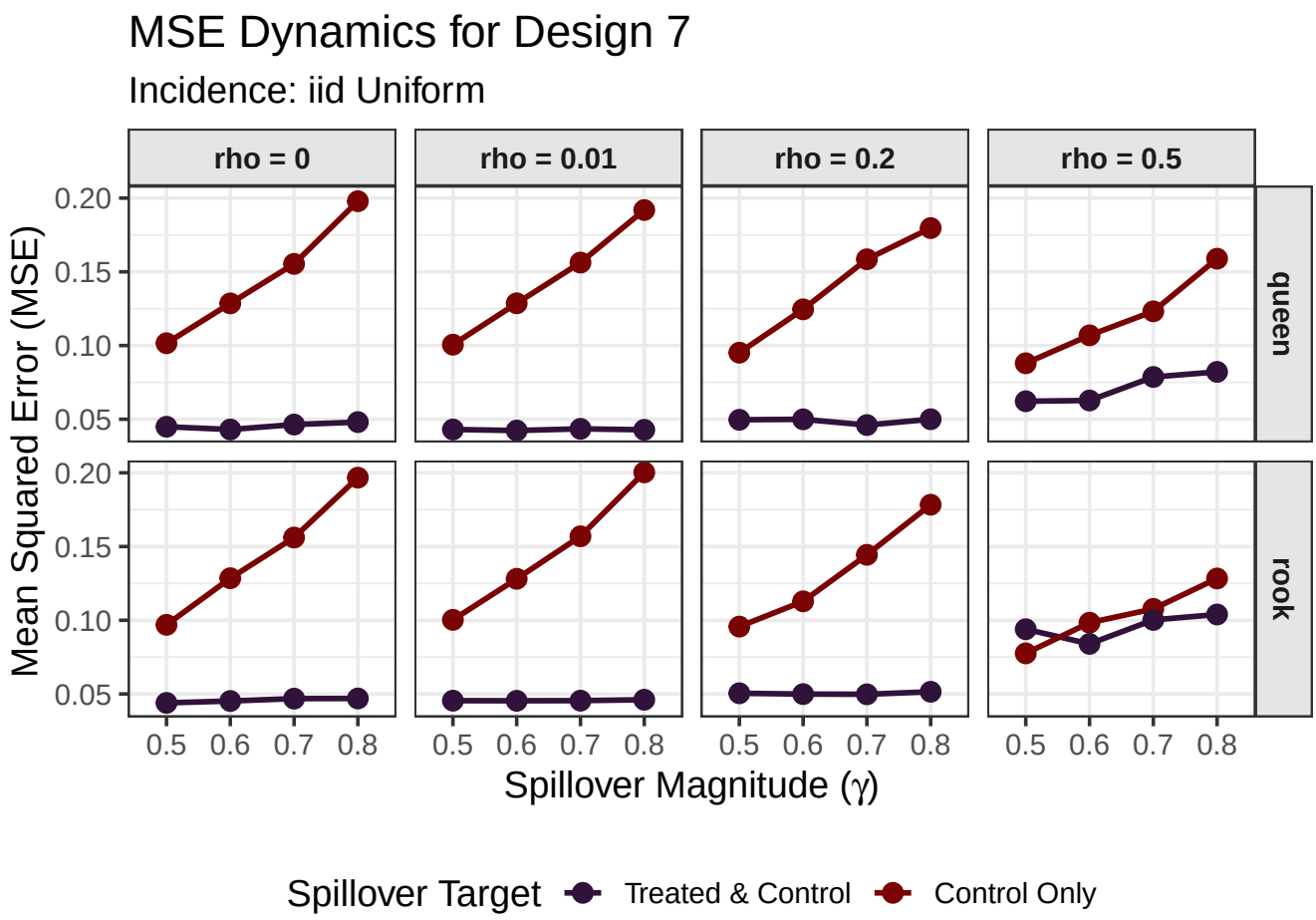

Figure 20: DIM: Per-design MSE line plots across gamma levels, for iid Uniform incidence. Each panel shows one design; lines represent different rho levels.

7.3 Incidence Mode Comparison

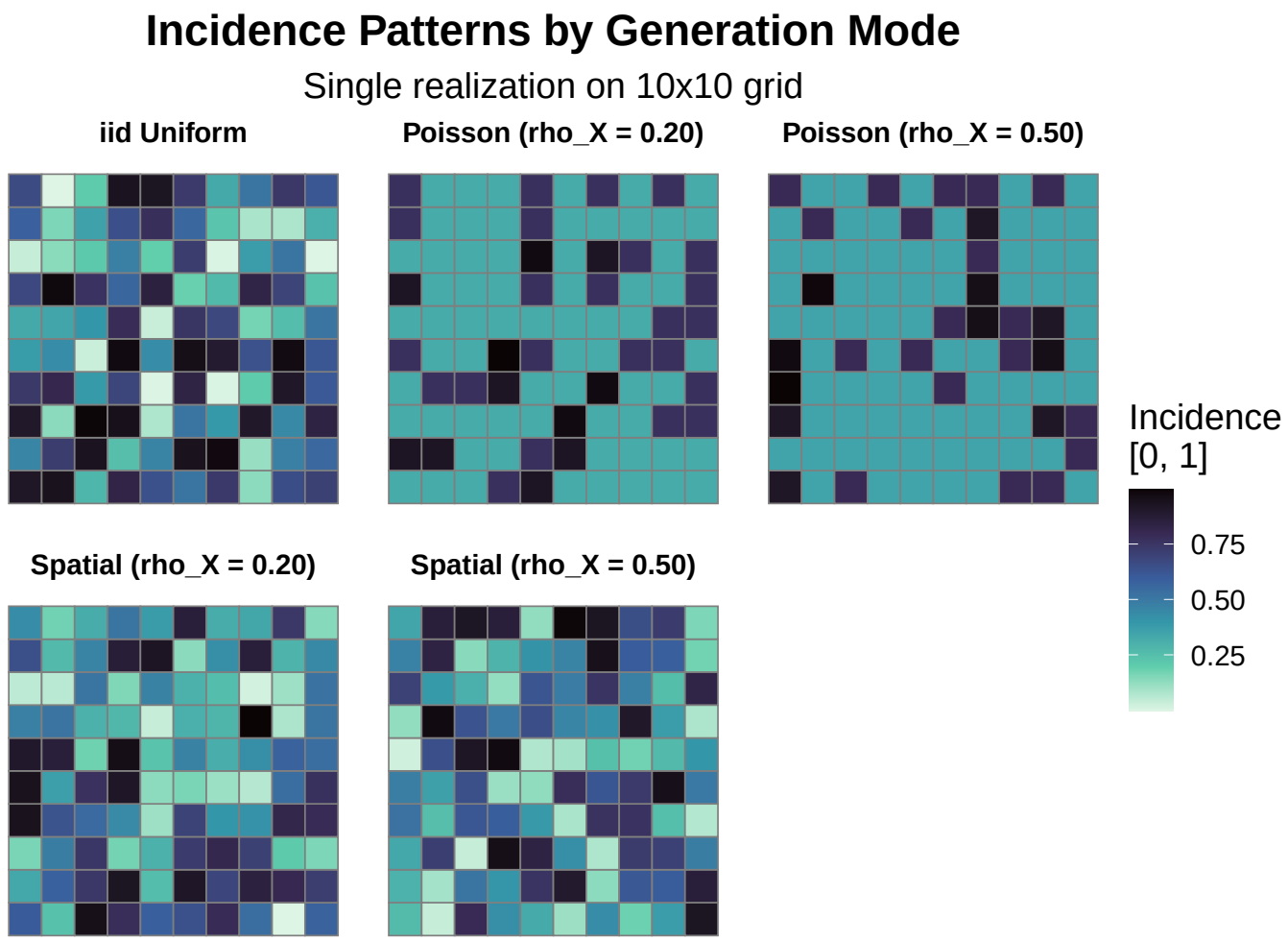

Figure 21: Incidence spatial patterns across the three generation modes for rho_X = 0.20 and 0.50. Color represents incidence (darker = higher burden). Spatial and Poisson modes show visible spatial clustering absent from iid.

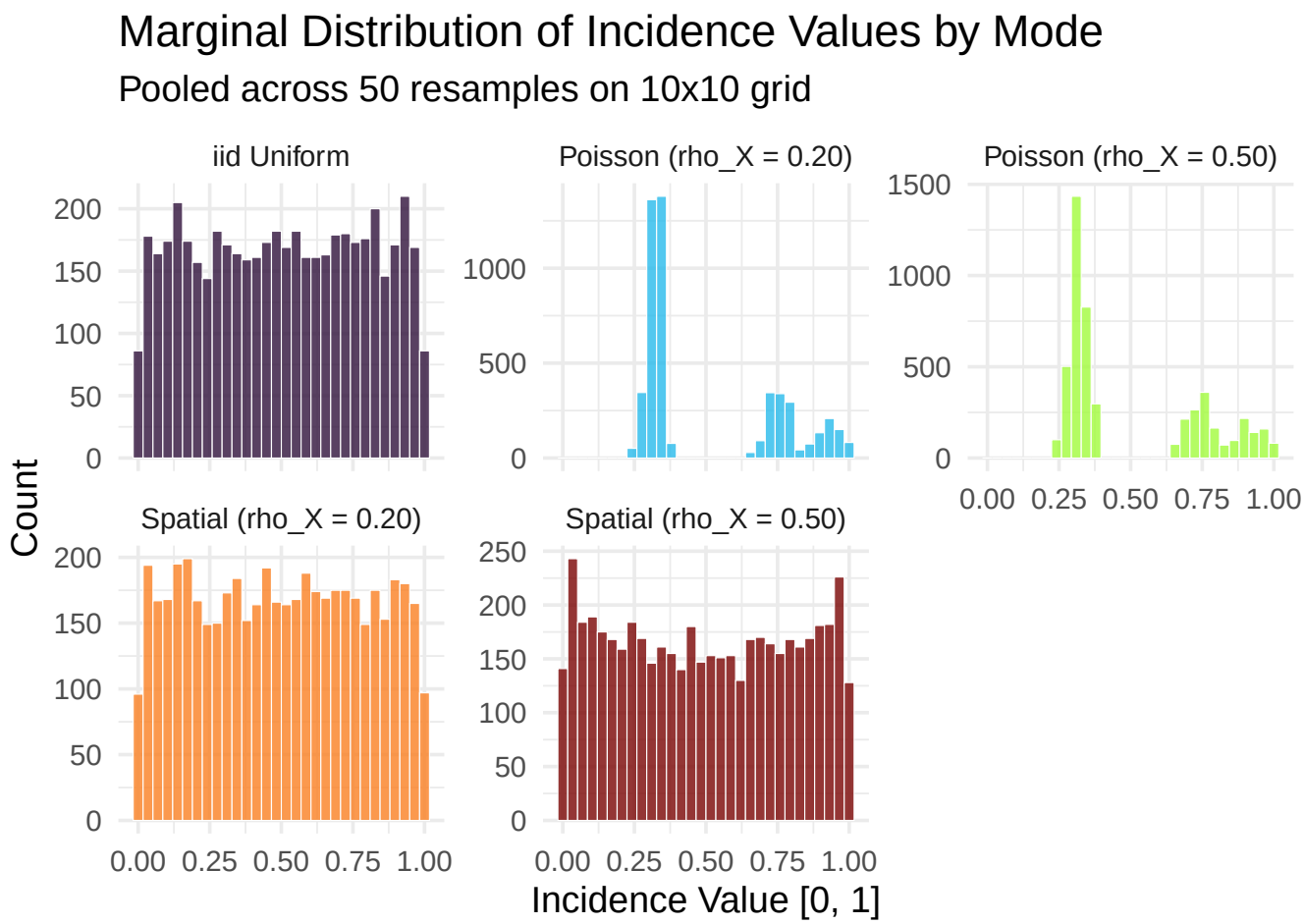

Figure 22: Marginal distribution of incidence values by mode, pooled across 50 resamples on the 10x10 grid. All modes are marginally Uniform on [0,1]; spatial structure manifests in spatial covariance, not the marginal.

---

8 Subgroup Results by Incidence Mode

The following tables report DIM results separately for each incidence configuration. Results are **never aggregated across incidence modes**, as each represents a fundamentally different assumption about prior data.

8.1 DIM: iid Uniform

Table 12: DIM robustness profile: iid Uniform

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 2	0.0434	0.0574	0.0897	0.1754	0.2812	0.1180
Design 3	0.0458	0.0605	0.0904	0.1306	0.3467	0.0701

Design 7	0.0423	0.0478	0.0945	0.1284	0.2003	0.0806
Design 5	0.0472	0.0575	0.0996	0.1933	0.2803	0.1358
Design 4	0.0775	0.0969	0.1287	0.2295	0.3213	0.1326
Design 1	0.0401	0.0962	0.2441	0.4764	0.7815	0.3801

8.2 DIM: Poisson (rho_X = 0.20)

Table 13: DIM robustness profile: Poisson (rho X = 0.20)

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 2	0.0447	0.0569	0.0706	0.0870	0.1132	0.0301
Design 3	0.0467	0.0620	0.0868	0.1307	0.3198	0.0686
Design 7	0.0432	0.0472	0.0905	0.1292	0.1990	0.0819
Design 5	0.0487	0.0568	0.1002	0.1890	0.2745	0.1322
Design 4	0.0792	0.0975	0.1343	0.2251	0.3123	0.1276
Design 1	0.0434	0.0964	0.2541	0.4971	0.7967	0.4007

8.3 DIM: Poisson (rho_X = 0.50)

Table 14: DIM robustness profile: Poisson (rho X = 0.50)

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 2	0.0538	0.0716	0.0836	0.1076	0.3393	0.0360
Design 3	0.0480	0.0638	0.0883	0.1318	0.3915	0.0680
Design 7	0.0427	0.0466	0.0902	0.1319	0.2000	0.0854
Design 5	0.0456	0.0571	0.0990	0.1910	0.2739	0.1339
Design 4	0.0795	0.0993	0.1334	0.2189	0.3146	0.1196
Design 1	0.0319	0.0804	0.2761	0.5354	0.7952	0.4550

8.4 DIM: Spatial (rho_X = 0.20)

Table 15: DIM robustness profile: Spatial (rho X = 0.20)

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 3	0.0472	0.0675	0.0891	0.1361	0.3716	0.0687
Design 7	0.0436	0.0468	0.0906	0.1270	0.2023	0.0802
Design 2	0.0415	0.0511	0.0941	0.1375	0.2165	0.0865
Design 5	0.0481	0.0593	0.1019	0.1877	0.2834	0.1284
Design 4	0.0776	0.0963	0.1293	0.2338	0.3097	0.1374
Design 1	0.0324	0.0850	0.2532	0.5083	0.7986	0.4234

8.5 DIM: Spatial (rho_X = 0.50)

Table 16: DIM robustness profile: Spatial (rho X = 0.50)

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE

Design 2	0.0396	0.0667	0.0879	0.1258	0.3491	0.0591
Design 7	0.0417	0.0472	0.0920	0.1280	0.2029	0.0808
Design 3	0.0481	0.0688	0.0927	0.1299	0.3553	0.0610
Design 5	0.0474	0.0561	0.1023	0.1911	0.2841	0.1350
Design 4	0.0774	0.0982	0.1343	0.2289	0.3254	0.1308
Design 1	0.0406	0.0991	0.2572	0.4919	0.7492	0.3927

9 Tau Sensitivity Analysis

The tau-sweep simulation (12,800 scenarios) swept the direct treatment effect τ across $\{0.8, 1.0, 1.5, 2.0, 3.0\}$. The primary scenario is $\tau = 1.0$; all standard results above use this slice. This section examines whether design rankings and statistical power change across effect sizes.

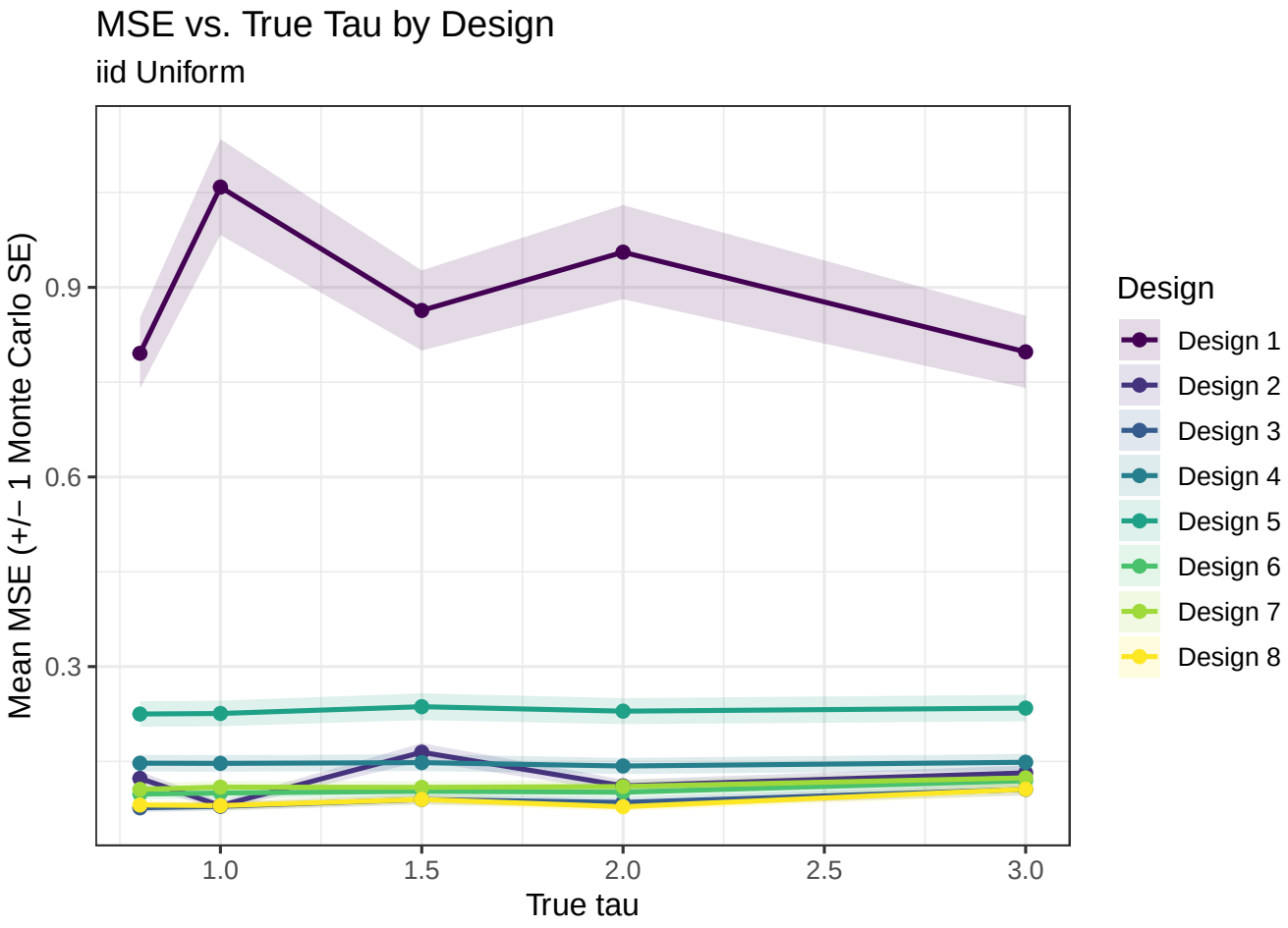

Figure 23: MLE MSE by design across tau values. Rankings should be stable across effect sizes; D1 (Checkerboard) is consistently worst.

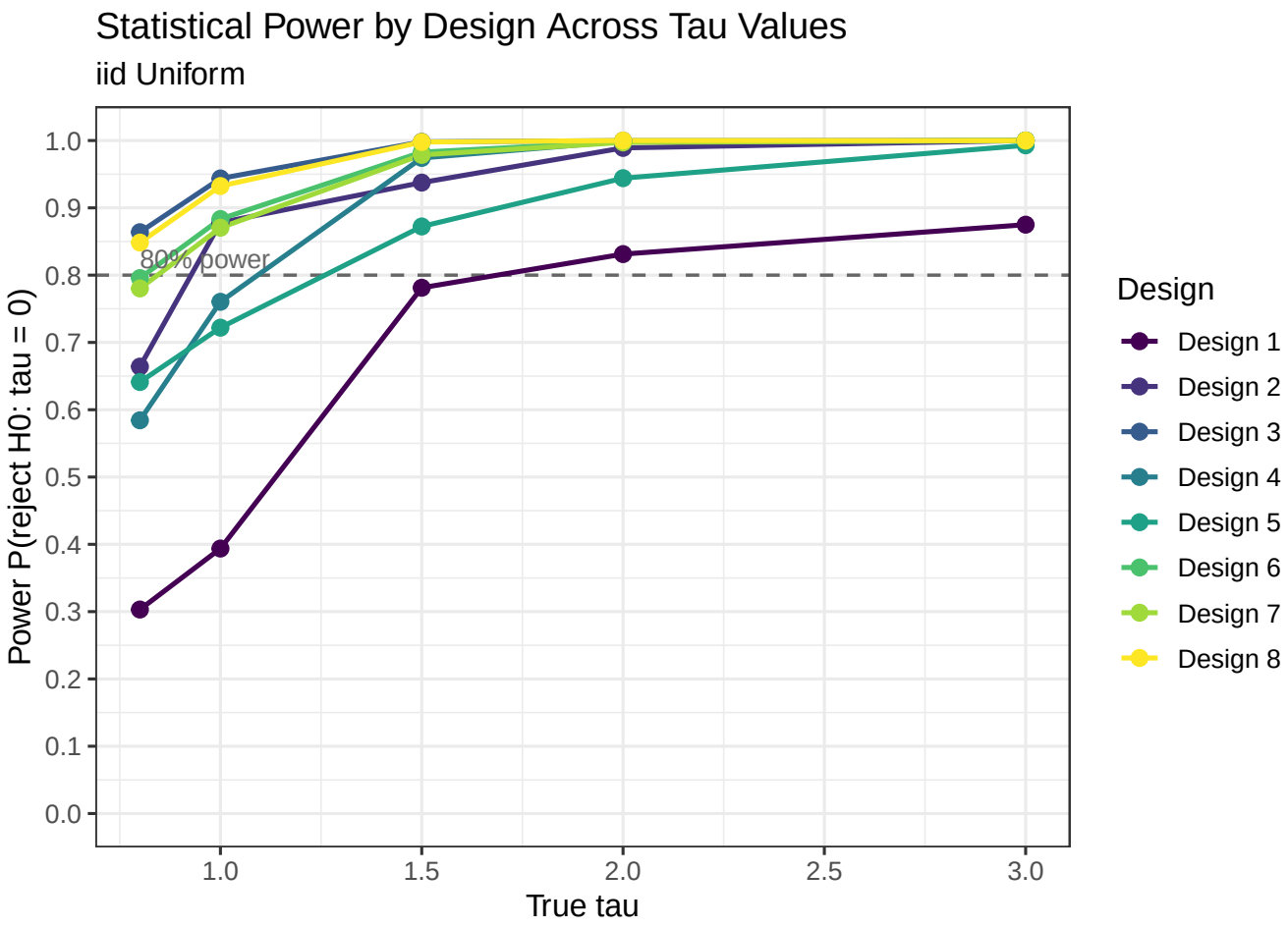

Figure 24: Statistical power P(reject H0: tau=0) by design across tau values. 80% power reference line shown. Designs with higher MSE require larger tau to achieve adequate power.

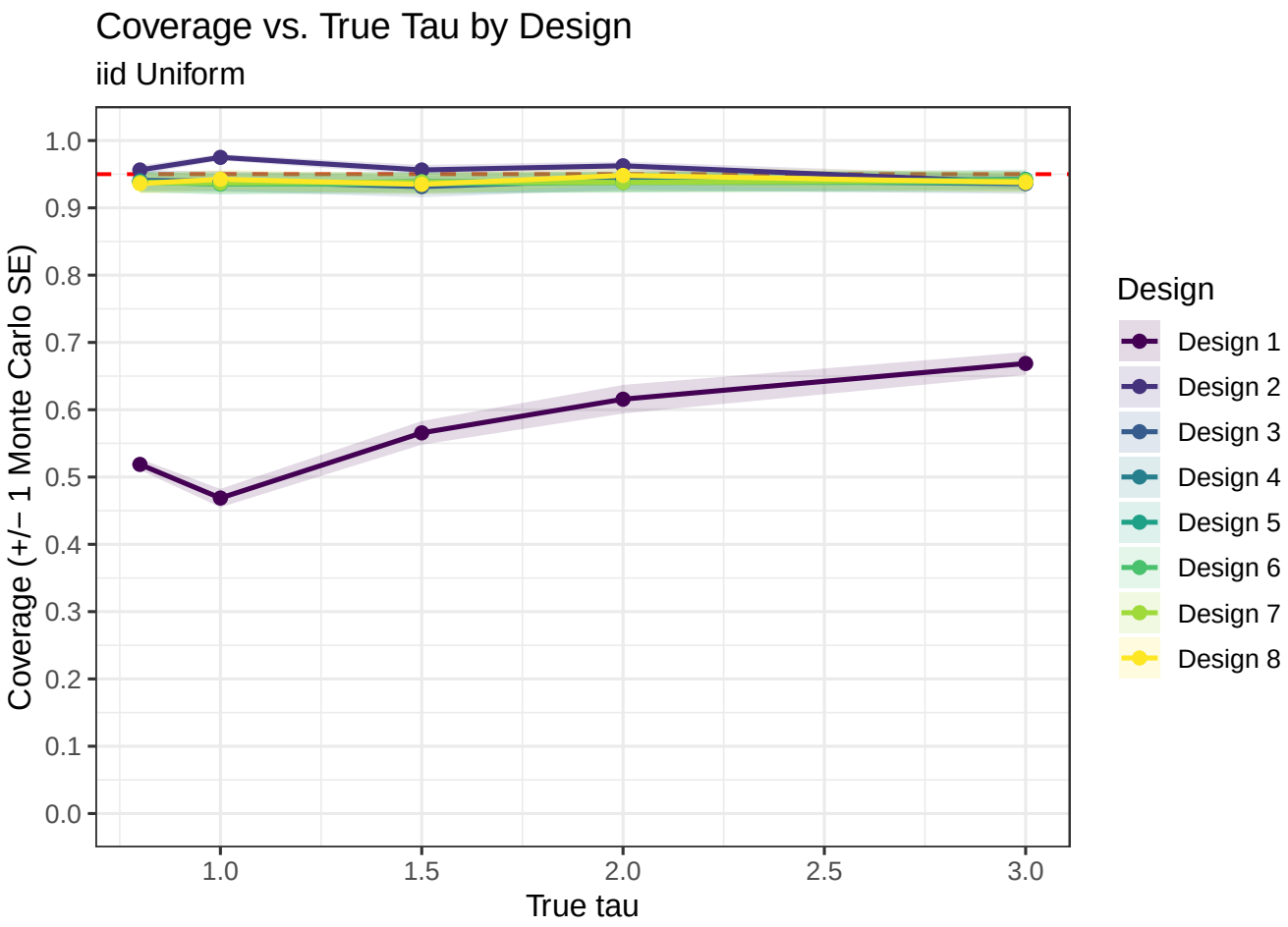

Figure 25: MLE coverage by design across tau values. Nominal 95% reference line shown. Coverage should be near-nominal for all designs except D1.

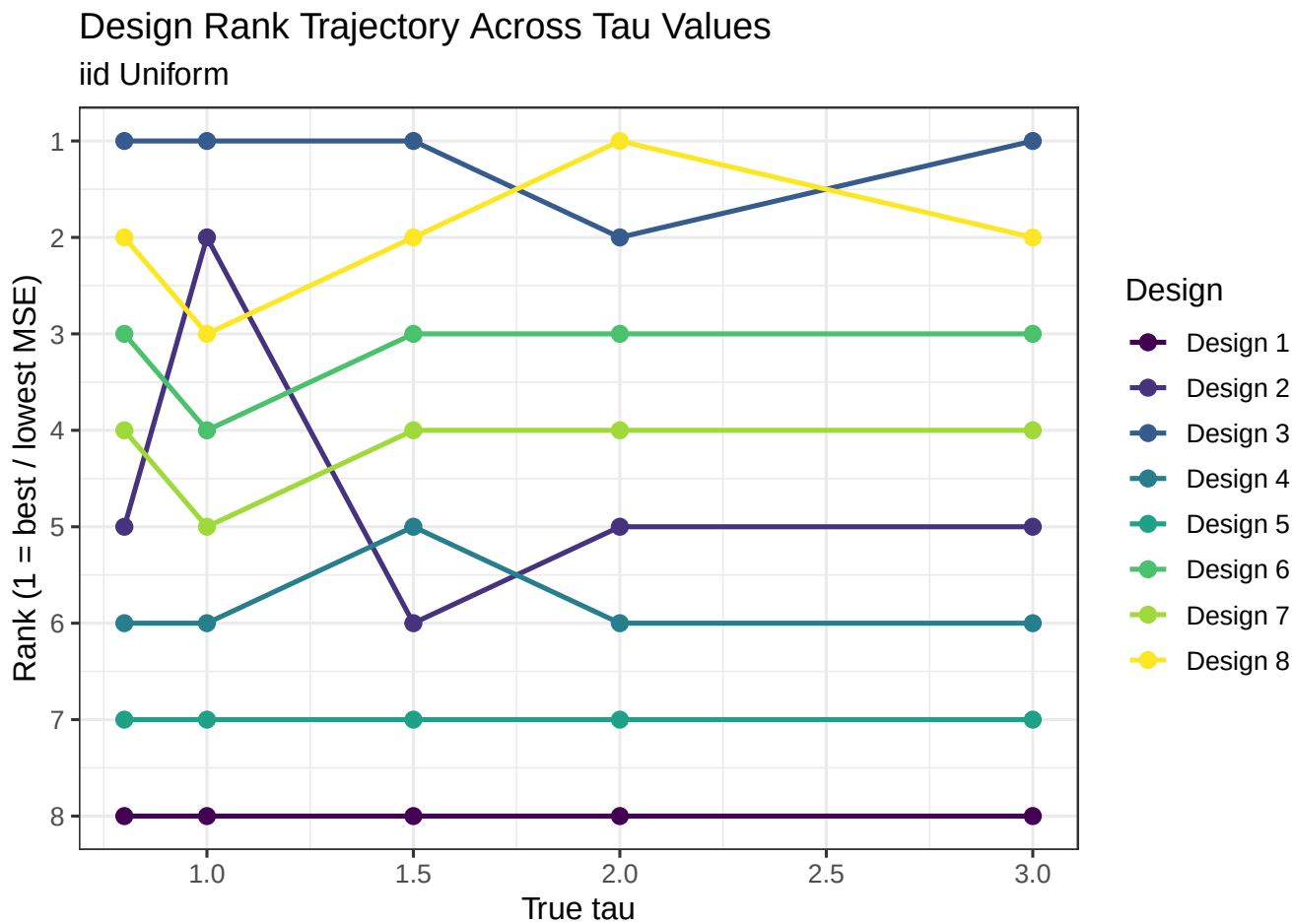

Figure 26: Design rank trajectories across tau values. Rank 1 = best MSE. Stable trajectories indicate robust design recommendations across effect sizes.

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10 Conclusions

10.1 Summary of Findings

The simulation study evaluated 8 treatment assignment designs across 2,560 MLE parameter scenarios (primary $\tau = 1.0$ slice), spanning three incidence generation modes, two contiguity structures, four spatial autocorrelation levels, four spillover magnitudes, and two spillover types. The full tau-sweep covers 12,800 scenarios across $\tau \in \{0.8, 1.0, 1.5, 2.0, 3.0\}$.

Key findings:

1. **Estimation method dominates coverage.** MLE achieves near-nominal 95% coverage for Designs 2–7, while DIM systematically under-covers (mean 72%). When spatial autocorrelation is present, a properly-specified spatial model is essential for valid inference.
2. **Design choice matters most under DIM.** Under MLE, coverage is uniformly high across most designs. Under DIM, the gap between best (Design 7, ~82%) and worst (Design 1, ~45%) designs is substantial. Researchers using simpler estimators should prefer stratified designs.

3. **Checkerboard is the consistently worst design.** Design 1's perfect spatial alternation creates systematic confounding when incidence is spatially structured, inflating bias and degrading coverage under both estimators.

4. **High Incidence Focus and Balanced Quartiles are robustly competitive.** Designs 2 and 7 rank consistently well across estimator, incidence mode, and parameter regimes. Design 7's incidence-quartile stratification directly addresses confounding from X.

5. **Saturation Quadrants excels under MLE.** Design 3 achieves the lowest average MSE under the oracle spatial model, likely because quadrant-level variation in treatment saturation provides rich spatial contrast for estimating ρ and τ jointly.

10.2 Limitations and Planned Extensions

- **Oracle MLE:** The MLE model uses the true spillover covariate. Non-oracle comparisons (`include_spill_covariate = FALSE`) remain as a planned extension.
- **Iteration count asymmetry:** DIM uses 2,500 iterations vs. 250 for MLE. MLE's apparent variance inflation at high-ρ may partly reflect this imbalance.
- **Equal populations:** Poisson mode uses $P_i = 1{,}000$ for all clusters. Heterogeneous population sizes (log-normal) would more closely match NC county data.
- **Grid sensitivity:** All results use $G = 10$ ($N = 100$). Sensitivity to $G = 8$ or $G = 15$ has not been evaluated.
- **Tau sensitivity:** τ was swept across $\{0.8, 1.0, 1.5, 2.0, 3.0\}$ in the completed tau-sweep run (12,800 scenarios). See Section 9 for power and rank-stability results across effect sizes.

---

11 References

- Mirzaei, A. et al. (2019). Sudden unexpected death rates. *(SUD base rate 35/100,000)*
- Gan, W. et al. (2019). County-level Poisson regression for SUD mortality.
- Watson, K. et al. (AHA). Census tract-level spatial analysis of SUD.
- LeSage, J. & Pace, R.K. (2009). *Introduction to Spatial Econometrics.* CRC Press.
- Bivand, R., et al. `spdep` R package.
- Bivand, R. & Piras, G. `spatialreg` R package.
- R packages used: `dplyr`, `tidyr`, `ggplot2`, `viridis`, `knitr`, `kableExtra`, `digest`

---

12 Appendix: Result File Locations

Table 17: Output files in results/ directory

File	Contents
sim_results_DIM_combined_20260304_195321.rds	All 1,920 DIM scenarios
sim_results_MLE_combined_20260305_150742.rds	All 1,920 MLE scenarios
sim_results_DIM_iid_*.rds / _spatial_*.rds / _poisson_*.rds	Per-incidence-mode DIM splits
sim_results_MLE_iid_*.rds / _spatial_*.rds / _poisson_*.rds	Per-incidence-mode MLE splits
DIM_combined_*.pdf (5 files)	Per-config 8-plot PDFs (DIM)
MLE_combined_*.pdf (5 files)	Per-config 8-plot PDFs (MLE)

DIM_combined_incidence_overview.pdf	Incidence heatmaps + distributions
MLE_combined_incidence_overview.pdf	Incidence heatmaps + distributions (MLE)
00_mathematical_specification.pdf	Mathematical theory document
07_results_summary.pdf	This report

All files are located in: `code/OutcomeIncidenceDesign/ModularIncidenceSim/results/`